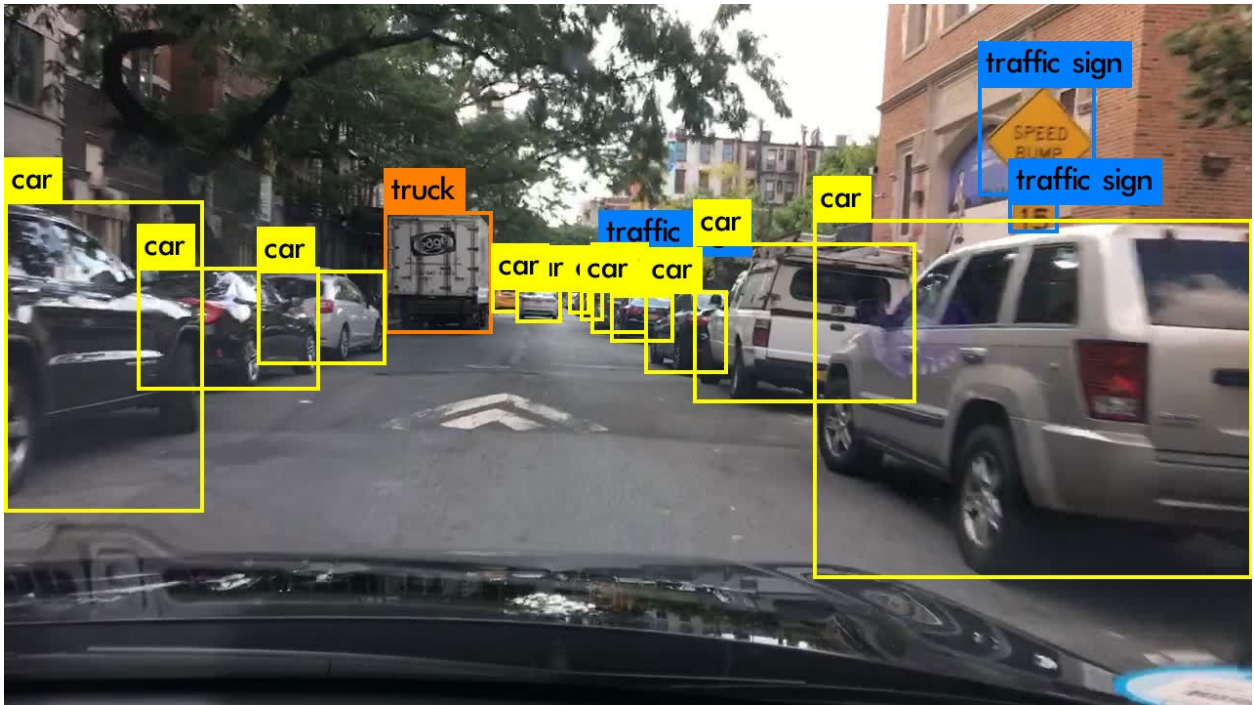


Autonomous Navigation - YOLO

prageethM



```
import os
import pandas as pd
import shutil
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import cv2

from IPython.display import Video
from ultralytics import YOLO

dataset_path = "/kaggle/input/udacity-self-driving-car-dataset/data"
csv_files = []
txt_files = []
for root, dirs, files in os.walk(dataset_path):
    for file in files:
        if file.endswith(".csv"):
            csv_files.append(os.path.join(root, file))
        if file.endswith(".txt"):
            txt_files.append(os.path.join(root, file))

print("CSV Files Found:", csv_files)
print("TXT Files Found:", txt_files)
```

```
CSV Files Found:
['/kaggle/input/udacity-self-driving-car-dataset/data/export/_annotations.csv']
TXT Files Found: []
```

```
csv_source_path =
"/kaggle/input/udacity-self-driving-car-dataset/data/export/_annotations.csv"
```

```
txt_source_path =
"/kaggle/input/udacity-self-driving-car-dataset/README.roboflow.txt"
```

```
csv_download_path = "/kaggle/working/"
```

```
txt_download_path = "/kaggle/working/"
```

```
try:
    shutil.copy(csv_source_path, csv_download_path)
    print(f"CSV file copied to: {csv_download_path}")
except FileNotFoundError:
    print("CSV file not found.")
```

```
try:
    shutil.copy(txt_source_path, txt_download_path)
    print(f"TXT file copied to: {txt_download_path}")
except FileNotFoundError:
    print("TXT file not found.")
```

```
CSV file copied to: /kaggle/working/
```

```
TXT file copied to: /kaggle/working/
```

```
csv_path = "/kaggle/working/_annotations.csv"
```

```
df = pd.read_csv(csv_path)
```

```
print("CSV File Structure:")
```

```
print(df.head())
```

```
CSV File Structure:
```

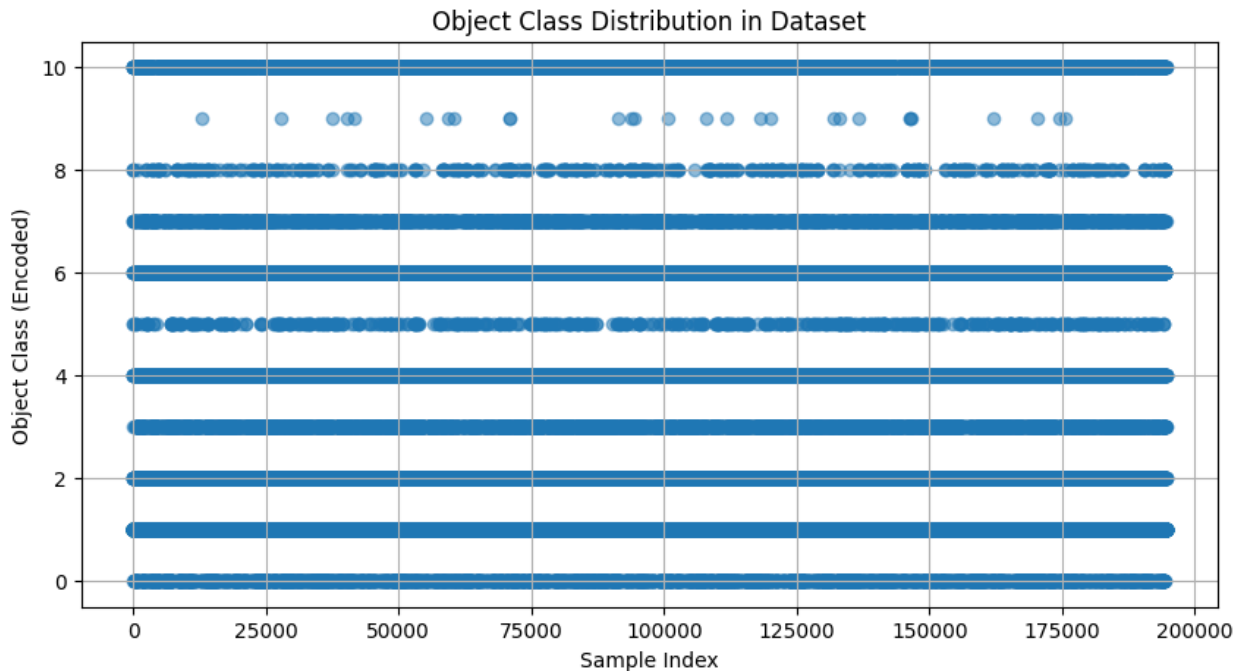
	filename	width	height	\
0	1478900859981702684_jpg.rf.6830635c7d919747563...	512	512	
1	1478900859981702684_jpg.rf.6830635c7d919747563...	512	512	
2	1478900859981702684_jpg.rf.6830635c7d919747563...	512	512	
3	1478900859981702684_jpg.rf.6830635c7d919747563...	512	512	
4	1478900859981702684_jpg.rf.6830635c7d919747563...	512	512	

	class	xmin	ymin	xmax	ymax
0	car	291	247	370	331
1	pedestrian	270	235	293	321
2	car	0	266	13	327
3	car	25	258	106	304
4	car	111	259	135	289

```
print(df.columns)
```

```
Index(['filename', 'width', 'height', 'class', 'xmin', 'ymin', 'xmax', 'ymax'], dtype='object')
```

```
plt.figure(figsize=(10, 5))
plt.scatter(range(len(df)), df["class"].astype("category").cat.codes,
            alpha = 0.5)
plt.xlabel("Sample Index")
plt.ylabel("Object Class (Encoded)")
plt.title("Object Class Distribution in Dataset")
plt.grid(True)
plt.show()
```



```
!pip install ultralytics opencv-python
```

Collecting ultralytics

```
  Downloading ultralytics-8.3.85-py3-none-any.whl.metadata (35 kB)
Requirement already satisfied: opencv-python in
/usr/local/lib/python3.10/dist-packages (4.10.0.84)
Requirement already satisfied: numpy<=2.1.1,>=1.23.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (1.26.4)
Requirement already satisfied: matplotlib>=3.3.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (3.7.5)
Requirement already satisfied: pillow>=7.1.2 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (11.0.0)
Requirement already satisfied: pyyaml>=5.3.1 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (6.0.2)
Requirement already satisfied: requests>=2.23.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (2.32.3)
Requirement already satisfied: scipy>=1.4.1 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (1.13.1)
Requirement already satisfied: torch>=1.8.0 in
```

```
/usr/local/lib/python3.10/dist-packages (from ultralytics)
(2.5.1+cu121)
Requirement already satisfied: torchvision>=0.9.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics)
(0.20.1+cu121)
Requirement already satisfied: tqdm>=4.64.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (4.67.1)
Requirement already satisfied: psutil in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: py-cpuinfo in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (9.0.0)
Requirement already satisfied: pandas>=1.1.4 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (2.2.3)
Requirement already satisfied: seaborn>=0.11.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (0.12.2)
Collecting ultralytics-thop>=2.0.0 (from ultralytics)
  Downloading ultralytics_thop-2.0.14-py3-none-any.whl.metadata (9.4
kB)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (1.3.1)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (4.55.3)
Requirement already satisfied: kiwisolver>=1.0.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (1.4.7)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (24.2)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (3.2.0)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (2.9.0.post0)
Requirement already satisfied: mkl_fft in
/usr/local/lib/python3.10/dist-packages (from numpy<=2.1.1,>=1.23.0-
>ultralytics) (1.3.8)
Requirement already satisfied: mkl_random in
/usr/local/lib/python3.10/dist-packages (from numpy<=2.1.1,>=1.23.0-
>ultralytics) (1.2.4)
Requirement already satisfied: mkl_umath in
/usr/local/lib/python3.10/dist-packages (from numpy<=2.1.1,>=1.23.0-
>ultralytics) (0.1.1)
Requirement already satisfied: mkl in /usr/local/lib/python3.10/dist-
```

packages (from numpy<=2.1.1,>=1.23.0->ultralitics) (2025.0.1)
Requirement already satisfied: tbb4py in
/usr/local/lib/python3.10/dist-packages (from numpy<=2.1.1,>=1.23.0->ultralitics) (2022.0.0)
Requirement already satisfied: mkl-service in
/usr/local/lib/python3.10/dist-packages (from numpy<=2.1.1,>=1.23.0->ultralitics) (2.4.1)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.10/dist-packages (from pandas>=1.1.4->ultralitics) (2025.1)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.10/dist-packages (from pandas>=1.1.4->ultralitics) (2025.1)
Requirement already satisfied: charset-normalizer<4,>=2 in
/usr/local/lib/python3.10/dist-packages (from requests>=2.23.0->ultralitics) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in
/usr/local/lib/python3.10/dist-packages (from requests>=2.23.0->ultralitics) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.10/dist-packages (from requests>=2.23.0->ultralitics) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.10/dist-packages (from requests>=2.23.0->ultralitics) (2025.1.31)
Requirement already satisfied: filelock in
/usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralitics) (3.17.0)
Requirement already satisfied: typing-extensions>=4.8.0 in
/usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralitics) (4.12.2)
Requirement already satisfied: networkx in
/usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralitics) (3.4.2)
Requirement already satisfied: jinja2 in
/usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralitics) (3.1.4)
Requirement already satisfied: fsspec in
/usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralitics) (2024.12.0)
Requirement already satisfied: sympy==1.13.1 in
/usr/local/lib/python3.10/dist-packages (from torch>=1.8.0->ultralitics) (1.13.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/usr/local/lib/python3.10/dist-packages (from sympy==1.13.1->torch>=1.8.0->ultralitics) (1.3.0)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.10/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralitics) (1.17.0)

```
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.10/dist-packages (from jinja2->torch>=1.8.0-
>ultralytics) (3.0.2)
Requirement already satisfied: intel-openmp>=2024 in
/usr/local/lib/python3.10/dist-packages (from mkl-
>numpy<=2.1.1,>=1.23.0->ultralytics) (2024.2.0)
Requirement already satisfied: tbb==2022.* in
/usr/local/lib/python3.10/dist-packages (from mkl-
>numpy<=2.1.1,>=1.23.0->ultralytics) (2022.0.0)
Requirement already satisfied: tcmlib==1.* in
/usr/local/lib/python3.10/dist-packages (from tbb==2022.*->mkl-
>numpy<=2.1.1,>=1.23.0->ultralytics) (1.2.0)
Requirement already satisfied: intel-cmplr-lib-rt in
/usr/local/lib/python3.10/dist-packages (from mkl_umath-
>numpy<=2.1.1,>=1.23.0->ultralytics) (2024.2.0)
Requirement already satisfied: intel-cmplr-lib-ur==2024.2.0 in
/usr/local/lib/python3.10/dist-packages (from intel-openmp>=2024->mkl-
>numpy<=2.1.1,>=1.23.0->ultralytics) (2024.2.0)
Downloading ultralytics-8.3.85-py3-none-any.whl (922 kB)
922.2/922.2 kB 17.5 MB/s eta
0:00:0000:01
```

```
yolo_labels_dir = "/kaggle/working/yolo_labels"
os.makedirs(yolo_labels_dir, exist_ok=True)
for index, row in df.iterrows():
    filename = row["filename"].replace(".jpg", ".txt")
    label_path = os.path.join(yolo_labels_dir, filename)

    width, height = row["width"], row["height"]
    x_center = (row["xmin"] + row["xmax"]) / 2 / width
    y_center = (row["ymin"] + row["ymax"]) / 2 / height
    bbox_width = (row["xmax"] - row["xmin"]) / width
    bbox_height = (row["ymax"] - row["ymin"]) / height

    with open(label_path, "w") as f:
        f.write(f"0 {x_center} {y_center} {bbox_width} {bbox_height}\n")

images_dir = "/kaggle/working/yolo_images"
os.makedirs(images_dir, exist_ok=True)
image_folder =
"/kaggle/input/udacity-self-driving-car-dataset/data/export"

for index, row in df.iterrows():
    img_path = os.path.join(image_folder, row["filename"])
    if os.path.exists(img_path):
        shutil.copy(img_path, images_dir)

dataset_path = "/kaggle/working/yolo_dataset"
images_path = os.path.join(dataset_path, "images")
```

```

labels_path = os.path.join(dataset_path, "labels")
os.makedirs(images_path, exist_ok=True)
os.makedirs(labels_path, exist_ok=True)

print("Dataset folders created successfully.")

Dataset folders created successfully.

csv_path = "/kaggle/working/_annotations.csv"
df = pd.read_csv(csv_path)
source_image_folder = "/kaggle/input/udacity-self-driving-car-
dataset/data/export"
for index, row in df.iterrows():
    img_filename = row["filename"]
    txt_filename = img_filename.replace(".jpg", ".txt")
    img_src_path = os.path.join(source_image_folder, img_filename)
    img_dest_path = os.path.join(images_path, img_filename)
    if os.path.exists(img_src_path):
        shutil.copy(img_src_path, img_dest_path)

    label_dest_path = os.path.join(labels_path, txt_filename)
    width, height = row["width"], row["height"]
    x_center = (row["xmin"] + row["xmax"]) / 2 / width
    y_center = (row["ymin"] + row["ymax"]) / 2 / height
    bbox_width = (row["xmax"] - row["xmin"]) / width
    bbox_height = (row["ymax"] - row["ymin"]) / height

    with open(label_dest_path, "w") as f:
        f.write(f"0 {x_center} {y_center} {bbox_width} {bbox_height}\n")
print("Images and labels moved successfully.")

Images and labels moved successfully.

yaml_path = "/kaggle/working/yolo_dataset/data.yaml"

with open(yaml_path, "w") as f:
    f.write(f"""
path: /kaggle/working/yolo_dataset
train: images
val: images
nc: 1
names: ['obstacle']
""")
print("Dataset configuration file created successfully.")

Dataset configuration file created successfully.

model = YOLO("yolov8n.pt")
model.train(data = "/kaggle/working/yolo_dataset/data.yaml", epochs =
5, imgsz = 512, batch = 8, name = "yolo_v8_model")

```

Creating new Ultralytics Settings v0.0.6 file ☐
View Ultralytics Settings with 'yolo settings' or at
'/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings
runs_dir=path/to/dir'. For help see
<https://docs.ultralytics.com/quickstart/#ultralytics-settings>.
Downloading
[https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8n](https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8n.pt)
.pt to 'yolov8n.pt'...

100%|██████████| 6.25M/6.25M [00:00<00:00, 78.4MB/s]

Ultralytics 8.3.85 ☐ Python-3.10.12 torch-2.5.1+cu121 CUDA:0 (Tesla
T4, 15095MiB)

engine/trainer: task=detect, mode=train, model=yolov8n.pt,
data=/kaggle/working/yolo_dataset/data.yaml, epochs=5, time=None,
patience=100, batch=8, imgsz=512, save=True, save_period=-1,
cache=False, device=None, workers=8, project=None, name=yolo_v8_model,
exist_ok=False, pretrained=True, optimizer=auto, verbose=True, seed=0,
deterministic=True, single_cls=False, rect=False, cos_lr=False,
close_mosaic=10, resume=False, amp=True, fraction=1.0, profile=False,
freeze=None, multi_scale=False, overlap_mask=True, mask_ratio=4,
dropout=0.0, val=True, split=val, save_json=False, save_hybrid=False,
conf=None, iou=0.7, max_det=300, half=False, dnn=False, plots=True,
source=None, vid_stride=1, stream_buffer=False, visualize=False,
augment=False, agnostic_nms=False, classes=None, retina_masks=False,
embed=None, show=False, save_frames=False, save_txt=False,
save_conf=False, save_crop=False, show_labels=True, show_conf=True,
show_boxes=True, line_width=None, format=torchscript, keras=False,
optimize=False, int8=False, dynamic=False, simplify=True, opset=None,
workspace=None, nms=False, lr0=0.01, lrf=0.01, momentum=0.937,
weight_decay=0.0005, warmup_epochs=3.0, warmup_momentum=0.8,
warmup_bias_lr=0.1, box=7.5, cls=0.5, dfl=1.5, pose=12.0, kobj=1.0,
nbs=64, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, degrees=0.0, translate=0.1,
scale=0.5, shear=0.0, perspective=0.0, flipud=0.0,fliplr=0.5,
bgr=0.0, mosaic=1.0, mixup=0.0, copy_paste=0.0, copy_paste_mode=flip,
auto_augment=randaugument, erasing=0.4, crop_fraction=1.0, cfg=None,
tracker=botsort.yaml, save_dir=runs/detect/yolo_v8_model
Downloading <https://ultralytics.com/assets/Arial.ttf> to
'/root/.config/Ultralytics/Arial.ttf'...

100%|██████████| 755k/755k [00:00<00:00, 17.6MB/s]

Overriding model.yaml nc=80 with nc=1

	from	n	params	module
arguments				
0	-1	1	464	ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]				
1	-1	1	4672	ultralytics.nn.modules.conv.Conv

2	[16, 32, 3, 2]	-1	1	7360	ultralytics.nn.modules.block.C2f
3	[32, 32, 1, True]	-1	1	18560	ultralytics.nn.modules.conv.Conv
4	[32, 64, 3, 2]	-1	2	49664	ultralytics.nn.modules.block.C2f
5	[64, 64, 2, True]	-1	1	73984	ultralytics.nn.modules.conv.Conv
6	[64, 128, 3, 2]	-1	2	197632	ultralytics.nn.modules.block.C2f
7	[128, 128, 2, True]	-1	1	295424	ultralytics.nn.modules.conv.Conv
8	[128, 256, 3, 2]	-1	1	460288	ultralytics.nn.modules.block.C2f
9	[256, 256, 1, True]	-1	1	164608	
	ultralytics.nn.modules.block.SPPF				[256, 256, 5]
10		-1	1	0	
	torch.nn.modules.upsampling.Upsample				[None, 2, 'nearest']
11		[-1, 6]	1	0	
	ultralytics.nn.modules.conv.Concat				[1]
12		-1	1	148224	ultralytics.nn.modules.block.C2f
	[384, 128, 1]				
13		-1	1	0	
	torch.nn.modules.upsampling.Upsample				[None, 2, 'nearest']
14		[-1, 4]	1	0	
	ultralytics.nn.modules.conv.Concat				[1]
15		-1	1	37248	ultralytics.nn.modules.block.C2f
	[192, 64, 1]				
16		-1	1	36992	ultralytics.nn.modules.conv.Conv
	[64, 64, 3, 2]				
17		[-1, 12]	1	0	
	ultralytics.nn.modules.conv.Concat				[1]
18		-1	1	123648	ultralytics.nn.modules.block.C2f
	[192, 128, 1]				
19		-1	1	147712	ultralytics.nn.modules.conv.Conv
	[128, 128, 3, 2]				
20		[-1, 9]	1	0	
	ultralytics.nn.modules.conv.Concat				[1]
21		-1	1	493056	ultralytics.nn.modules.block.C2f
	[384, 256, 1]				
22		[15, 18, 21]	1	751507	
	ultralytics.nn.modules.head.Detect				[1, [64, 128, 256]]

Model summary: 129 layers, 3,011,043 parameters, 3,011,027 gradients, 8.2 GFLOPs

Transferred 319/355 items from pretrained weights

TensorBoard: Start with 'tensorboard --logdir runs/detect/yolo_v8_model', view at http://localhost:6006/

Freezing layer 'model.22.dfl.conv.weight'

AMP: running Automatic Mixed Precision (AMP) checks...

Downloading

<https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt> to 'yolo11n.pt'...

100%|██████████| 5.35M/5.35M [00:00<00:00, 73.5MB/s]

AMP: checks passed ☐

train: Scanning /kaggle/working/yolo_dataset/labels... 26300 images, 0 backgrounds, 0 corrupt: 100%|██████████| 26300/26300 [00:20<00:00, 1265.87it/s]

train: New cache created: /kaggle/working/yolo_dataset/labels.cache
augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, num_output_channels=3, method='weighted_average'), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

/usr/local/lib/python3.10/dist-packages/albumentations/__init__.py:24:
UserWarning: A new version of Albumentations is available: 2.0.5 (you have 1.4.20). Upgrade using: pip install -U albumentations. To disable automatic update checks, set the environment variable NO_ALBUMENTATIONS_UPDATE to 1.

check_for_updates()

val: Scanning /kaggle/working/yolo_dataset/labels.cache... 26300 images, 0 backgrounds, 0 corrupt: 100%|██████████| 26300/26300 [00:00<?, ?it/s]

Plotting labels to runs/detect/yolo_v8_model/labels.jpg...

optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum' automatically...

optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)

TensorBoard: model graph visualization added ☐

Image sizes 512 train, 512 val

Using 2 dataloader workers

Logging results to runs/detect/yolo_v8_model

Starting training for 5 epochs...

	Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size						

1/5	0.736G	1.737	2.274	1.288	4
512: 100%	██████████	3288/3288	[04:44<00:00, 11.54it/s]		
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	1644/1644	[02:14<00:00, 12.21it/s]	
	all	26300	26300	0.342	0.4
0.307	0.168				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
2/5	0.84G	1.707	1.785	1.3	3
512: 100%	██████████	3288/3288	[04:34<00:00, 11.99it/s]		
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	1644/1644	[02:14<00:00, 12.23it/s]	
	all	26300	26300	0.516	0.461
0.448	0.247				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
3/5	0.82G	1.653	1.648	1.265	7
512: 100%	██████████	3288/3288	[04:30<00:00, 12.18it/s]		
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	1644/1644	[02:13<00:00, 12.32it/s]	
	all	26300	26300	0.551	0.529
0.512	0.292				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
4/5	0.824G	1.584	1.512	1.232	7
512: 100%	██████████	3288/3288	[04:28<00:00, 12.26it/s]		
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	1644/1644	[02:12<00:00, 12.41it/s]	
	all	26300	26300	0.603	0.573
0.586	0.34				

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances
Size					
5/5	0.816G	1.496	1.37	1.188	5
512: 100%	██████████	3288/3288	[04:28<00:00, 12.27it/s]		
	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%	██████████	1644/1644	[02:12<00:00, 12.38it/s]	
	all	26300	26300	0.658	0.607
0.641	0.38				

```
5 epochs completed in 0.568 hours.
Optimizer stripped from runs/detect/yolo_v8_model/weights/last.pt,
6.2MB
Optimizer stripped from runs/detect/yolo_v8_model/weights/best.pt,
6.2MB

Validating runs/detect/yolo_v8_model/weights/best.pt...
Ultralytics 8.3.85 | Python-3.10.12 torch-2.5.1+cu121 CUDA:0 (Tesla
T4, 15095MiB)
Model summary (fused): 72 layers, 3,005,843 parameters, 0 gradients,
8.1 GFLOPs
```

	Class	Images	Instances	Box(P	R
mAP50	mAP50-95): 100%		1644/1644	[02:08<00:00,	12.84it/s]
	all	26300	26300	0.659	0.606
0.641	0.38				

```
/usr/local/lib/python3.10/dist-packages/matplotlib/colors.py:721:
RuntimeWarning: invalid value encountered in less
  xa[xa < 0] = -1
/usr/local/lib/python3.10/dist-packages/matplotlib/colors.py:721:
RuntimeWarning: invalid value encountered in less
  xa[xa < 0] = -1
```

```
Speed: 0.1ms preprocess, 1.3ms inference, 0.0ms loss, 0.7ms
postprocess per image
Results saved to runs/detect/yolo_v8_model
```

```
ultralytics.utils.metrics.DetMetrics object with attributes:
```

```
ap_class_index: array([0])
box: ultralytics.utils.metrics.Metric object
confusion_matrix: <ultralytics.utils.metrics.ConfusionMatrix object at
0x7b7000780520>
curves: ['Precision-Recall(B)', 'F1-Confidence(B)', 'Precision-
Confidence(B)', 'Recall-Confidence(B)']
curves_results: [[array([
0, 0.001001, 0.002002,
0.003003, 0.004004, 0.005005, 0.006006, 0.007007,
0.008008, 0.009009, 0.01001, 0.011011, 0.012012,
0.013013, 0.014014, 0.015015, 0.016016, 0.017017,
0.018018, 0.019019, 0.02002, 0.021021, 0.022022,
0.023023,
0.024024, 0.025025, 0.026026, 0.027027,
0.028028, 0.029029, 0.03003, 0.031031, 0.032032,
0.033033, 0.034034, 0.035035, 0.036036, 0.037037,
0.038038, 0.039039, 0.04004, 0.041041, 0.042042,
0.043043, 0.044044, 0.045045, 0.046046, 0.047047,
0.048048, 0.049049, 0.05005, 0.051051,
0.052052, 0.053053, 0.054054, 0.055055, 0.056056,
0.057057, 0.058058, 0.059059, 0.06006, 0.061061,
```

0.062062,	0.063063,	0.064064,	0.065065,	0.066066,
0.067067,	0.068068,	0.069069,	0.07007,	0.071071,
	0.072072,	0.073073,	0.074074,	0.075075,
0.076076,	0.077077,	0.078078,	0.079079,	0.08008,
0.081081,	0.082082,	0.083083,	0.084084,	0.085085,
0.086086,	0.087087,	0.088088,	0.089089,	0.09009,
0.091091,	0.092092,	0.093093,	0.094094,	0.095095,
	0.096096,	0.097097,	0.098098,	0.099099,
0.1001,	0.1011,	0.1021,	0.1031,	0.1041,
0.10511,	0.10611,	0.10711,	0.10811,	0.10911,
0.11011,	0.11111,	0.11211,	0.11311,	0.11411,
0.11512,	0.11612,	0.11712,	0.11812,	0.11912,
	0.12012,	0.12112,	0.12212,	0.12312,
0.12412,	0.12513,	0.12613,	0.12713,	0.12813,
0.12913,	0.13013,	0.13113,	0.13213,	0.13313,
0.13413,	0.13514,	0.13614,	0.13714,	0.13814,
0.13914,	0.14014,	0.14114,	0.14214,	0.14314,
	0.14414,	0.14515,	0.14615,	0.14715,
0.14815,	0.14915,	0.15015,	0.15115,	0.15215,
0.15315,	0.15415,	0.15516,	0.15616,	0.15716,
0.15816,	0.15916,	0.16016,	0.16116,	0.16216,
0.16316,	0.16416,	0.16517,	0.16617,	0.16717,
	0.16817,	0.16917,	0.17017,	0.17117,
0.17217,	0.17317,	0.17417,	0.17518,	0.17618,
0.17718,	0.17818,	0.17918,	0.18018,	0.18118,
0.18218,	0.18318,	0.18418,	0.18519,	0.18619,
0.18719,	0.18819,	0.18919,	0.19019,	0.19119,
	0.19219,	0.19319,	0.19419,	0.1952,
0.1962,	0.1972,	0.1982,	0.1992,	0.2002,
0.2012,	0.2022,	0.2032,	0.2042,	0.20521,
0.20621,	0.20721,	0.20821,	0.20921,	0.21021,
0.21121,	0.21221,	0.21321,	0.21421,	0.21522,
	0.21622,	0.21722,	0.21822,	0.21922,
0.22022,	0.22122,	0.22222,	0.22322,	0.22422,
0.22523,	0.22623,	0.22723,	0.22823,	0.22923,
0.23023,	0.23123,	0.23223,	0.23323,	0.23423,
0.23524,	0.23624,	0.23724,	0.23824,	0.23924,
	0.24024,	0.24124,	0.24224,	0.24324,
0.24424,	0.24525,	0.24625,	0.24725,	0.24825,
0.24925,	0.25025,	0.25125,	0.25225,	0.25325,
0.25425,	0.25526,	0.25626,	0.25726,	0.25826,
0.25926,	0.26026,	0.26126,	0.26226,	0.26326,
	0.26426,	0.26527,	0.26627,	0.26727,
0.26827,	0.26927,	0.27027,	0.27127,	0.27227,
0.27327,	0.27427,	0.27528,	0.27628,	0.27728,
0.27828,	0.27928,	0.28028,	0.28128,	0.28228,
0.28328,	0.28428,	0.28529,	0.28629,	0.28729,
	0.28829,	0.28929,	0.29029,	0.29129,
0.29229,	0.29329,	0.29429,	0.2953,	0.2963,

0.2973,	0.2983,	0.2993,	0.3003,	0.3013,
0.3023,	0.3033,	0.3043,	0.30531,	0.30631,
0.30731,	0.30831,	0.30931,	0.31031,	0.31131,
	0.31231,	0.31331,	0.31431,	0.31532,
0.31632,	0.31732,	0.31832,	0.31932,	0.32032,
0.32132,	0.32232,	0.32332,	0.32432,	0.32533,
0.32633,	0.32733,	0.32833,	0.32933,	0.33033,
0.33133,	0.33233,	0.33333,	0.33433,	0.33534,
	0.33634,	0.33734,	0.33834,	0.33934,
0.34034,	0.34134,	0.34234,	0.34334,	0.34434,
0.34535,	0.34635,	0.34735,	0.34835,	0.34935,
0.35035,	0.35135,	0.35235,	0.35335,	0.35435,
0.35536,	0.35636,	0.35736,	0.35836,	0.35936,
	0.36036,	0.36136,	0.36236,	0.36336,
0.36436,	0.36537,	0.36637,	0.36737,	0.36837,
0.36937,	0.37037,	0.37137,	0.37237,	0.37337,
0.37437,	0.37538,	0.37638,	0.37738,	0.37838,
0.37938,	0.38038,	0.38138,	0.38238,	0.38338,
	0.38438,	0.38539,	0.38639,	0.38739,
0.38839,	0.38939,	0.39039,	0.39139,	0.39239,
0.39339,	0.39439,	0.3954,	0.3964,	0.3974,
0.3984,	0.3994,	0.4004,	0.4014,	0.4024,
0.4034,	0.4044,	0.40541,	0.40641,	0.40741,
	0.40841,	0.40941,	0.41041,	0.41141,
0.41241,	0.41341,	0.41441,	0.41542,	0.41642,
0.41742,	0.41842,	0.41942,	0.42042,	0.42142,
0.42242,	0.42342,	0.42442,	0.42543,	0.42643,
0.42743,	0.42843,	0.42943,	0.43043,	0.43143,
	0.43243,	0.43343,	0.43443,	0.43544,
0.43644,	0.43744,	0.43844,	0.43944,	0.44044,
0.44144,	0.44244,	0.44344,	0.44444,	0.44545,
0.44645,	0.44745,	0.44845,	0.44945,	0.45045,
0.45145,	0.45245,	0.45345,	0.45445,	0.45546,
	0.45646,	0.45746,	0.45846,	0.45946,
0.46046,	0.46146,	0.46246,	0.46346,	0.46446,
0.46547,	0.46647,	0.46747,	0.46847,	0.46947,
0.47047,	0.47147,	0.47247,	0.47347,	0.47447,
0.47548,	0.47648,	0.47748,	0.47848,	0.47948,
	0.48048,	0.48148,	0.48248,	0.48348,
0.48448,	0.48549,	0.48649,	0.48749,	0.48849,
0.48949,	0.49049,	0.49149,	0.49249,	0.49349,
0.49449,	0.4955,	0.4965,	0.4975,	0.4985,
0.4995,	0.5005,	0.5015,	0.5025,	0.5035,
	0.5045,	0.50551,	0.50651,	0.50751,
0.50851,	0.50951,	0.51051,	0.51151,	0.51251,
0.51351,	0.51451,	0.51552,	0.51652,	0.51752,
0.51852,	0.51952,	0.52052,	0.52152,	0.52252,
0.52352,	0.52452,	0.52553,	0.52653,	0.52753,
	0.52853,	0.52953,	0.53053,	0.53153,

0.53253,	0.53353,	0.53453,	0.53554,	0.53654,
0.53754,	0.53854,	0.53954,	0.54054,	0.54154,
0.54254,	0.54354,	0.54454,	0.54555,	0.54655,
0.54755,	0.54855,	0.54955,	0.55055,	0.55155,
	0.55255,	0.55355,	0.55455,	0.55556,
0.55656,	0.55756,	0.55856,	0.55956,	0.56056,
0.56156,	0.56256,	0.56356,	0.56456,	0.56557,
0.56657,	0.56757,	0.56857,	0.56957,	0.57057,
0.57157,	0.57257,	0.57357,	0.57457,	0.57558,
	0.57658,	0.57758,	0.57858,	0.57958,
0.58058,	0.58158,	0.58258,	0.58358,	0.58458,
0.58559,	0.58659,	0.58759,	0.58859,	0.58959,
0.59059,	0.59159,	0.59259,	0.59359,	0.59459,
0.5956,	0.5966,	0.5976,	0.5986,	0.5996,
	0.6006,	0.6016,	0.6026,	0.6036,
0.6046,	0.60561,	0.60661,	0.60761,	0.60861,
0.60961,	0.61061,	0.61161,	0.61261,	0.61361,
0.61461,	0.61562,	0.61662,	0.61762,	0.61862,
0.61962,	0.62062,	0.62162,	0.62262,	0.62362,
	0.62462,	0.62563,	0.62663,	0.62763,
0.62863,	0.62963,	0.63063,	0.63163,	0.63263,
0.63363,	0.63463,	0.63564,	0.63664,	0.63764,
0.63864,	0.63964,	0.64064,	0.64164,	0.64264,
0.64364,	0.64464,	0.64565,	0.64665,	0.64765,
	0.64865,	0.64965,	0.65065,	0.65165,
0.65265,	0.65365,	0.65465,	0.65566,	0.65666,
0.65766,	0.65866,	0.65966,	0.66066,	0.66166,
0.66266,	0.66366,	0.66466,	0.66567,	0.66667,
0.66767,	0.66867,	0.66967,	0.67067,	0.67167,
	0.67267,	0.67367,	0.67467,	0.67568,
0.67668,	0.67768,	0.67868,	0.67968,	0.68068,
0.68168,	0.68268,	0.68368,	0.68468,	0.68569,
0.68669,	0.68769,	0.68869,	0.68969,	0.69069,
0.69169,	0.69269,	0.69369,	0.69469,	0.6957,
	0.6967,	0.6977,	0.6987,	0.6997,
0.7007,	0.7017,	0.7027,	0.7037,	0.7047,
0.70571,	0.70671,	0.70771,	0.70871,	0.70971,
0.71071,	0.71171,	0.71271,	0.71371,	0.71471,
0.71572,	0.71672,	0.71772,	0.71872,	0.71972,
	0.72072,	0.72172,	0.72272,	0.72372,
0.72472,	0.72573,	0.72673,	0.72773,	0.72873,
0.72973,	0.73073,	0.73173,	0.73273,	0.73373,
0.73473,	0.73574,	0.73674,	0.73774,	0.73874,
0.73974,	0.74074,	0.74174,	0.74274,	0.74374,
	0.74474,	0.74575,	0.74675,	0.74775,
0.74875,	0.74975,	0.75075,	0.75175,	0.75275,
0.75375,	0.75475,	0.75576,	0.75676,	0.75776,
0.75876,	0.75976,	0.76076,	0.76176,	0.76276,
0.76376,	0.76476,	0.76577,	0.76677,	0.76777,

	0.76877,	0.76977,	0.77077,	0.77177,	
0.77277,	0.77377,	0.77477,	0.77578,	0.77678,	
0.77778,	0.77878,	0.77978,	0.78078,	0.78178,	
0.78278,	0.78378,	0.78478,	0.78579,	0.78679,	
0.78779,	0.78879,	0.78979,	0.79079,	0.79179,	
	0.79279,	0.79379,	0.79479,	0.7958,	
0.7968,	0.7978,	0.7988,	0.7998,	0.8008,	
0.8018,	0.8028,	0.8038,	0.8048,	0.80581,	
0.80681,	0.80781,	0.80881,	0.80981,	0.81081,	
0.81181,	0.81281,	0.81381,	0.81481,	0.81582,	
	0.81682,	0.81782,	0.81882,	0.81982,	
0.82082,	0.82182,	0.82282,	0.82382,	0.82482,	
0.82583,	0.82683,	0.82783,	0.82883,	0.82983,	
0.83083,	0.83183,	0.83283,	0.83383,	0.83483,	
0.83584,	0.83684,	0.83784,	0.83884,	0.83984,	
	0.84084,	0.84184,	0.84284,	0.84384,	
0.84484,	0.84585,	0.84685,	0.84785,	0.84885,	
0.84985,	0.85085,	0.85185,	0.85285,	0.85385,	
0.85485,	0.85586,	0.85686,	0.85786,	0.85886,	
0.85986,	0.86086,	0.86186,	0.86286,	0.86386,	
	0.86486,	0.86587,	0.86687,	0.86787,	
0.86887,	0.86987,	0.87087,	0.87187,	0.87287,	
0.87387,	0.87487,	0.87588,	0.87688,	0.87788,	
0.87888,	0.87988,	0.88088,	0.88188,	0.88288,	
0.88388,	0.88488,	0.88589,	0.88689,	0.88789,	
	0.88889,	0.88989,	0.89089,	0.89189,	
0.89289,	0.89389,	0.89489,	0.8959,	0.8969,	
0.8979,	0.8989,	0.8999,	0.9009,	0.9019,	
0.9029,	0.9039,	0.9049,	0.90591,	0.90691,	
0.90791,	0.90891,	0.90991,	0.91091,	0.91191,	
	0.91291,	0.91391,	0.91491,	0.91592,	
0.91692,	0.91792,	0.91892,	0.91992,	0.92092,	
0.92192,	0.92292,	0.92392,	0.92492,	0.92593,	
0.92693,	0.92793,	0.92893,	0.92993,	0.93093,	
0.93193,	0.93293,	0.93393,	0.93493,	0.93594,	
	0.93694,	0.93794,	0.93894,	0.93994,	
0.94094,	0.94194,	0.94294,	0.94394,	0.94494,	
0.94595,	0.94695,	0.94795,	0.94895,	0.94995,	
0.95095,	0.95195,	0.95295,	0.95395,	0.95495,	
0.95596,	0.95696,	0.95796,	0.95896,	0.95996,	
	0.96096,	0.96196,	0.96296,	0.96396,	
0.96496,	0.96597,	0.96697,	0.96797,	0.96897,	
0.96997,	0.97097,	0.97197,	0.97297,	0.97397,	
0.97497,	0.97598,	0.97698,	0.97798,	0.97898,	
0.97998,	0.98098,	0.98198,	0.98298,	0.98398,	
	0.98498,	0.98599,	0.98699,	0.98799,	
0.98899,	0.98999,	0.99099,	0.99199,	0.99299,	
0.99399,	0.99499,	0.996,	0.997,	0.998,	
0.999,	1]], array([[	1,	1,	1,	1,

1,	1,	1,	1,	1,	1,
1,	1,	1,	1,	1,	1,
0.99751,	0.99323,	0.99129,	0.99129,	0.99129,	
0.99129,	0.99129,	0.98377,			
	0.98377,	0.98323,	0.98282,	0.98282,	
0.98282,	0.98282,	0.98282,	0.98282,	0.98278,	
0.98278,	0.98208,	0.98065,	0.97881,	0.97756,	
0.97756,	0.97756,	0.97691,	0.97691,	0.97691,	
0.97668,	0.97607,	0.97485,	0.97388,		
	0.97333,	0.97333,	0.97208,	0.97208,	
0.97208,	0.97038,	0.96965,	0.96877,	0.96877,	
0.96877,	0.96877,	0.96877,	0.96877,	0.96877,	
0.96877,	0.96712,	0.96704,	0.96704,	0.96704,	
0.96704,	0.96704,	0.96704,	0.96704,		
	0.96704,	0.96704,	0.96704,	0.96704,	
0.96704,	0.96704,	0.96704,	0.96704,	0.96704,	
0.96704,	0.96611,	0.96505,	0.9649,	0.96387,	
0.96356,	0.96351,	0.96299,	0.96235,	0.96143,	
0.95982,	0.95869,	0.95855,	0.95843,		
	0.95771,	0.95741,	0.95569,	0.95569,	
0.95551,	0.9549,	0.95453,	0.95448,	0.95433,	
0.95387,	0.95387,	0.95373,	0.95284,	0.95261,	
0.95162,	0.95143,	0.95143,	0.95099,	0.95086,	
0.95086,	0.95086,	0.95086,	0.95013,		
	0.94998,	0.94998,	0.94995,	0.94995,	
0.94911,	0.94809,	0.94777,	0.94697,	0.94626,	
0.9456,	0.94521,	0.94517,	0.94517,	0.94504,	
0.94352,	0.94323,	0.94323,	0.94323,	0.94323,	
0.94323,	0.94247,	0.94196,	0.94168,		
	0.9415,	0.94122,	0.94056,	0.94004,	
0.93992,	0.93979,	0.93953,	0.93842,	0.93842,	
0.93842,	0.93813,	0.93775,	0.93691,	0.93578,	
0.9354,	0.93533,	0.93522,	0.93503,	0.93503,	
0.93503,	0.93503,	0.93503,	0.93425,		
	0.93385,	0.93385,	0.93385,	0.93385,	
0.93385,	0.93357,	0.93357,	0.93357,	0.93357,	
0.9332,	0.9332,	0.9332,	0.9332,	0.93261,	
0.93134,	0.93112,	0.93076,	0.93056,	0.9293,	
0.92776,	0.92766,	0.92763,	0.92613,		
	0.92575,	0.92508,	0.92501,	0.92462,	
0.92405,	0.92353,	0.92312,	0.92273,	0.92246,	
0.92182,	0.92182,	0.92126,	0.92068,	0.92035,	
0.92005,	0.91932,	0.91918,	0.91829,	0.91796,	
0.91781,	0.91593,	0.91531,	0.91521,		
	0.91396,	0.91364,	0.91275,	0.91275,	
0.91275,	0.91172,	0.91131,	0.91125,	0.9106,	
0.91031,	0.90929,	0.90847,	0.90819,	0.90778,	
0.90739,	0.90723,	0.90701,	0.9067,	0.90606,	
0.90552,	0.90538,	0.9051,	0.90489,		

	0.90484,	0.90386,	0.9037,	0.90359,	
0.90321,	0.90285,	0.90283,	0.90246,	0.90246,	
0.90158,	0.90093,	0.89962,	0.89911,	0.89818,	
0.897,	0.89675,	0.89675,	0.89653,	0.89628,	
0.89607,	0.89575,	0.89524,	0.89453,		
	0.89426,	0.89363,	0.89285,	0.89217,	
0.89208,	0.89195,	0.89187,	0.89142,	0.89085,	
0.89052,	0.88977,	0.8891,	0.88861,	0.88848,	
0.88848,	0.88797,	0.88767,	0.8876,	0.8875,	
0.88703,	0.88664,	0.886,	0.88552,		
	0.8842,	0.88401,	0.88311,	0.88285,	
0.88269,	0.88206,	0.88175,	0.88065,	0.88018,	
0.87962,	0.87959,	0.87881,	0.87807,	0.87754,	
0.87691,	0.87593,	0.87551,	0.87474,	0.87407,	
0.87383,	0.87368,	0.87285,	0.8724,		
	0.87197,	0.87139,	0.87091,	0.87015,	
0.87,	0.86998,	0.86971,	0.86971,	0.86971,	0.86927,
0.86927,	0.86911,	0.86886,	0.86847,	0.86804,	
0.8672,	0.86695,	0.86658,	0.86595,	0.86565,	
0.8651,	0.86454,	0.86402,			
	0.86357,	0.86309,	0.86281,	0.86265,	
0.86212,	0.86186,	0.86116,	0.86092,	0.86081,	
0.86055,	0.85971,	0.85955,	0.85891,	0.85866,	
0.8581,	0.85758,	0.85685,	0.85629,	0.85615,	
0.85533,	0.85476,	0.8545,	0.85398,		
	0.85347,	0.85322,	0.85289,	0.85221,	
0.8514,	0.85111,	0.85076,	0.84944,	0.84873,	
0.84819,	0.8479,	0.84724,	0.84692,	0.84663,	
0.8465,	0.84644,	0.84594,	0.84532,	0.84503,	
0.84464,	0.84307,	0.84227,	0.84222,		
	0.8415,	0.84033,	0.83962,	0.83883,	
0.83791,	0.8378,	0.83697,	0.83661,	0.83596,	
0.83527,	0.8348,	0.83423,	0.83369,	0.83343,	
0.83288,	0.83259,	0.83177,	0.83177,	0.83092,	
0.83041,	0.83002,	0.8293,	0.82869,		
	0.82836,	0.82732,	0.82711,	0.82649,	
0.826,	0.82578,	0.82549,	0.82518,	0.8244,	
0.82396,	0.82341,	0.82267,	0.82178,	0.82114,	
0.8205,	0.82034,	0.82009,	0.8195,	0.81873,	
0.81806,	0.817,	0.8167,	0.81611,		
	0.81577,	0.81494,	0.81421,	0.81326,	
0.8124,	0.81213,	0.81121,	0.81048,	0.80949,	
0.80873,	0.80777,	0.80729,	0.80582,	0.8054,	
0.80484,	0.80447,	0.80401,	0.80358,	0.80323,	
0.80214,	0.80121,	0.80052,	0.80018,		
	0.79962,	0.79915,	0.79848,	0.7976,	
0.79725,	0.79645,	0.79578,	0.79501,	0.79441,	
0.79374,	0.79317,	0.79241,	0.79127,	0.7905,	
0.79003,	0.78871,	0.78828,	0.78685,	0.78609,	

0.78537,	0.785,	0.78419,	0.78315,	
	0.7827,	0.78227,	0.78098,	0.78074,
0.77988,	0.77955,	0.77874,	0.77762,	0.77676,
0.77631,	0.77558,	0.77524,	0.7748,	0.77421,
0.77351,	0.77335,	0.77293,	0.772,	0.77109,
0.77006,	0.76948,	0.76851,	0.76756,	
	0.76727,	0.76616,	0.76559,	0.76479,
0.76406,	0.76296,	0.76238,	0.76187,	0.76156,
0.76098,	0.75985,	0.7589,	0.758,	0.75771,
0.75739,	0.75694,	0.75569,	0.75452,	0.75415,
0.75283,	0.75214,	0.75092,	0.75072,	
	0.75025,	0.74928,	0.74885,	0.74838,
0.74808,	0.74729,	0.74648,	0.74611,	0.74539,
0.74507,	0.74418,	0.744,	0.74297,	0.74148,
0.7408,	0.73928,	0.73897,	0.73854,	0.7379,
0.7375,	0.73702,	0.73633,	0.73583,	
	0.73474,	0.73379,	0.73291,	0.73215,
0.73161,	0.73072,	0.72916,	0.72861,	0.72773,
0.72632,	0.72565,	0.72465,	0.72353,	0.72314,
0.72152,	0.7206,	0.71928,	0.71759,	0.71675,
0.71555,	0.71472,	0.71397,	0.71385,	
	0.7131,	0.71155,	0.71042,	0.70872,
0.70705,	0.70609,	0.70487,	0.70411,	0.70335,
0.70283,	0.70238,	0.70115,	0.70053,	0.69985,
0.69843,	0.69741,	0.69589,	0.69497,	0.6941,
0.69358,	0.69257,	0.69159,	0.69006,	
	0.68898,	0.68887,	0.68812,	0.68706,
0.68634,	0.68551,	0.68451,	0.68353,	0.6823,
0.68044,	0.67982,	0.67815,	0.67712,	0.67642,
0.67575,	0.67511,	0.67441,	0.67344,	0.67224,
0.67168,	0.67052,	0.66953,	0.66841,	
	0.66723,	0.66622,	0.6653,	0.66394,
0.66297,	0.66143,	0.66038,	0.65938,	0.65811,
0.6562,	0.65507,	0.6539,	0.65263,	0.65139,
0.64964,	0.64832,	0.64678,	0.6459,	0.64328,
0.64232,	0.64136,	0.64031,	0.63822,	
	0.63764,	0.6361,	0.6349,	0.63427,
0.63341,	0.6322,	0.63076,	0.62959,	0.62818,
0.62679,	0.6254,	0.6242,	0.6228,	0.62118,
0.62008,	0.61889,	0.61768,	0.61609,	0.61436,
0.61338,	0.61161,	0.61084,	0.6098,	
	0.60784,	0.60654,	0.60555,	0.60471,
0.60338,	0.6019,	0.60025,	0.59793,	0.59636,
0.59544,	0.59362,	0.59185,	0.59038,	0.58909,
0.58846,	0.58748,	0.58586,	0.58412,	0.58158,
0.57971,	0.57838,	0.5771,	0.57501,	
	0.57354,	0.57242,	0.57107,	0.56834,
0.56682,	0.56542,	0.564,	0.56278,	0.56136,
0.5596,	0.55754,	0.55549,	0.5533,	0.55114,

0.54928,	0.54819,	0.54675,	0.54471,	0.54301,
0.54163,	0.54031,	0.53829,	0.53632,	
	0.53505,	0.5331,	0.53055,	0.52918,
0.52745,	0.52535,	0.52315,	0.52178,	0.52015,
0.51788,	0.51561,	0.51411,	0.51278,	0.51083,
0.50884,	0.5072,	0.50631,	0.50416,	0.50295,
0.5014,	0.49952,	0.49761,	0.49591,	
	0.4938,	0.49216,	0.48986,	0.48837,
0.48598,	0.48441,	0.48283,	0.48156,	0.48048,
0.47848,	0.47677,	0.47479,	0.47239,	0.47074,
0.46832,	0.46651,	0.46455,	0.46319,	0.4613,
0.45869,	0.45617,	0.45481,	0.45283,	
	0.45075,	0.44944,	0.44721,	0.44535,
0.44384,	0.44158,	0.43945,	0.43767,	0.4356,
0.43445,	0.43247,	0.43078,	0.42916,	0.42662,
0.42512,	0.42277,	0.42079,	0.41893,	0.41702,
0.4144,	0.413,	0.41117,	0.41015,	
	0.40829,	0.40634,	0.40445,	0.40244,
0.40078,	0.39854,	0.39615,	0.39366,	0.39175,
0.39009,	0.38779,	0.38582,	0.38413,	0.38206,
0.38019,	0.37821,	0.37673,	0.37388,	0.37208,
0.37009,	0.36691,	0.36481,	0.36201,	
	0.36057,	0.35802,	0.35549,	0.35396,
0.35225,	0.3491,	0.34664,	0.3447,	0.34267,
0.34053,	0.33744,	0.33529,	0.33341,	0.33061,
0.32813,	0.32643,	0.32286,	0.32042,	0.31659,
0.31369,	0.31195,	0.30965,	0.30788,	
	0.30439,	0.3021,	0.3,	0.29631,
0.29366,	0.29094,	0.28728,	0.28587,	0.28328,
0.2812,	0.27927,	0.27754,	0.27569,	0.27366,
0.27021,	0.26848,	0.26632,	0.26303,	0.26018,
0.25818,	0.25643,	0.25387,	0.25175,	
	0.24934,	0.2477,	0.24587,	0.24385,
0.2412,	0.23883,	0.23587,	0.23321,	0.22981,
0.22772,	0.22506,	0.22251,	0.21943,	0.21671,
0.21432,	0.21167,	0.20864,	0.20523,	0.20201,
0.20023,	0.19765,	0.1958,	0.19378,	
	0.19219,	0.19042,	0.18861,	0.18615,
0.18361,	0.18142,	0.17843,	0.17628,	0.17378,
0.17076,	0.168,	0.16584,	0.1635,	0.16086,
0.15812,	0.15661,	0.15445,	0.15131,	0.14912,
0.14678,	0.14463,	0.1419,	0.13922,	
	0.13685,	0.13481,	0.13286,	0.13039,
0.12758,	0.12532,	0.12278,	0.12063,	0.11868,
0.11689,	0.11515,	0.11341,	0.11116,	0.10942,
0.1079,	0.10545,	0.10361,	0.10209,	0.099138,
0.097639,	0.095943,	0.094174,	0.09223,	
	0.089738,	0.088547,	0.086344,	0.084541,
0.083133,	0.081345,	0.079321,	0.078052,	0.07644,

```
0.074638,    0.073029,    0.071625,    0.069824,    0.067804,
0.065664,    0.063741,    0.062447,    0.060774,    0.059375,
0.057613,    0.055932,    0.054373,    0.052676,
0.04395,     0.05103,     0.048672,    0.047185,    0.045501,
0.042768,    0.041325,    0.039827,    0.03821,     0.036327,
0.035181,    0.033769,    0.03225,     0.030749,    0.029483,
0.02903,     0.028576,    0.028122,    0.027669,    0.027215,
0.026762,    0.026308,    0.025854,
0.025401,    0.024947,    0.024494,    0.02404,
0.023586,    0.023133,    0.022679,    0.022226,    0.021772,
0.021319,    0.020865,    0.020411,    0.019958,    0.019504,
0.019051,    0.018597,    0.018143,    0.01769,     0.017236,
0.016783,    0.016329,    0.015876,    0.015422,
0.014968,    0.014515,    0.014061,    0.013608,
0.013154,    0.0127,     0.012247,    0.011793,    0.01134,
0.010886,    0.010432,    0.0099789,   0.0095253,   0.0090717,
0.0086181,   0.0081646,   0.007711,    0.0072574,   0.0068038,
0.0063502,   0.0058966,   0.005443,    0.0049895,
0.0045359,   0.0040823,   0.0036287,    0.0031751,
0.0027215,   0.0022679,   0.0018143,    0.0013608,   0.00090717,
0.00045359, 0]]), 'Recall', 'Precision'],
[array([
0, 0.001001, 0.002002, 0.003003,
0.004004, 0.005005, 0.006006, 0.007007, 0.008008,
0.009009, 0.01001, 0.011011, 0.012012, 0.013013,
0.014014, 0.015015, 0.016016, 0.017017, 0.018018,
0.019019, 0.02002, 0.021021, 0.022022, 0.023023,
0.024024, 0.025025, 0.026026, 0.027027,
0.028028, 0.029029, 0.03003, 0.031031, 0.032032,
0.033033, 0.034034, 0.035035, 0.036036, 0.037037,
0.038038, 0.039039, 0.04004, 0.041041, 0.042042,
0.043043, 0.044044, 0.045045, 0.046046, 0.047047,
0.048048, 0.049049, 0.05005, 0.051051,
0.052052, 0.053053, 0.054054, 0.055055, 0.056056,
0.057057, 0.058058, 0.059059, 0.06006, 0.061061,
0.062062, 0.063063, 0.064064, 0.065065, 0.066066,
0.067067, 0.068068, 0.069069, 0.07007, 0.071071,
0.072072, 0.073073, 0.074074, 0.075075,
0.076076, 0.077077, 0.078078, 0.079079, 0.08008,
0.081081, 0.082082, 0.083083, 0.084084, 0.085085,
0.086086, 0.087087, 0.088088, 0.089089, 0.09009,
0.091091, 0.092092, 0.093093, 0.094094, 0.095095,
0.096096, 0.097097, 0.098098, 0.099099,
0.1001, 0.1011, 0.1021, 0.1031, 0.1041,
0.10511, 0.10611, 0.10711, 0.10811, 0.10911,
0.11011, 0.11111, 0.11211, 0.11311, 0.11411,
0.11512, 0.11612, 0.11712, 0.11812, 0.11912,
0.12012, 0.12112, 0.12212, 0.12312,
0.12412, 0.12513, 0.12613, 0.12713, 0.12813,
0.12913, 0.13013, 0.13113, 0.13213, 0.13313,
```

0.13413,	0.13514,	0.13614,	0.13714,	0.13814,
0.13914,	0.14014,	0.14114,	0.14214,	0.14314,
	0.14414,	0.14515,	0.14615,	0.14715,
0.14815,	0.14915,	0.15015,	0.15115,	0.15215,
0.15315,	0.15415,	0.15516,	0.15616,	0.15716,
0.15816,	0.15916,	0.16016,	0.16116,	0.16216,
0.16316,	0.16416,	0.16517,	0.16617,	0.16717,
	0.16817,	0.16917,	0.17017,	0.17117,
0.17217,	0.17317,	0.17417,	0.17518,	0.17618,
0.17718,	0.17818,	0.17918,	0.18018,	0.18118,
0.18218,	0.18318,	0.18418,	0.18519,	0.18619,
0.18719,	0.18819,	0.18919,	0.19019,	0.19119,
	0.19219,	0.19319,	0.19419,	0.1952,
0.1962,	0.1972,	0.1982,	0.1992,	0.2002,
0.2012,	0.2022,	0.2032,	0.2042,	0.20521,
0.20621,	0.20721,	0.20821,	0.20921,	0.21021,
0.21121,	0.21221,	0.21321,	0.21421,	0.21522,
	0.21622,	0.21722,	0.21822,	0.21922,
0.22022,	0.22122,	0.22222,	0.22322,	0.22422,
0.22523,	0.22623,	0.22723,	0.22823,	0.22923,
0.23023,	0.23123,	0.23223,	0.23323,	0.23423,
0.23524,	0.23624,	0.23724,	0.23824,	0.23924,
	0.24024,	0.24124,	0.24224,	0.24324,
0.24424,	0.24525,	0.24625,	0.24725,	0.24825,
0.24925,	0.25025,	0.25125,	0.25225,	0.25325,
0.25425,	0.25526,	0.25626,	0.25726,	0.25826,
0.25926,	0.26026,	0.26126,	0.26226,	0.26326,
	0.26426,	0.26527,	0.26627,	0.26727,
0.26827,	0.26927,	0.27027,	0.27127,	0.27227,
0.27327,	0.27427,	0.27528,	0.27628,	0.27728,
0.27828,	0.27928,	0.28028,	0.28128,	0.28228,
0.28328,	0.28428,	0.28529,	0.28629,	0.28729,
	0.28829,	0.28929,	0.29029,	0.29129,
0.29229,	0.29329,	0.29429,	0.2953,	0.2963,
0.2973,	0.2983,	0.2993,	0.3003,	0.3013,
0.3023,	0.3033,	0.3043,	0.30531,	0.30631,
0.30731,	0.30831,	0.30931,	0.31031,	0.31131,
	0.31231,	0.31331,	0.31431,	0.31532,
0.31632,	0.31732,	0.31832,	0.31932,	0.32032,
0.32132,	0.32232,	0.32332,	0.32432,	0.32533,
0.32633,	0.32733,	0.32833,	0.32933,	0.33033,
0.33133,	0.33233,	0.33333,	0.33433,	0.33534,
	0.33634,	0.33734,	0.33834,	0.33934,
0.34034,	0.34134,	0.34234,	0.34334,	0.34434,
0.34535,	0.34635,	0.34735,	0.34835,	0.34935,
0.35035,	0.35135,	0.35235,	0.35335,	0.35435,
0.35536,	0.35636,	0.35736,	0.35836,	0.35936,
	0.36036,	0.36136,	0.36236,	0.36336,
0.36436,	0.36537,	0.36637,	0.36737,	0.36837,
0.36937,	0.37037,	0.37137,	0.37237,	0.37337,

0.37437,	0.37538,	0.37638,	0.37738,	0.37838,
0.37938,	0.38038,	0.38138,	0.38238,	0.38338,
	0.38438,	0.38539,	0.38639,	0.38739,
0.38839,	0.38939,	0.39039,	0.39139,	0.39239,
0.39339,	0.39439,	0.3954,	0.3964,	0.3974,
0.3984,	0.3994,	0.4004,	0.4014,	0.4024,
0.4034,	0.4044,	0.40541,	0.40641,	0.40741,
	0.40841,	0.40941,	0.41041,	0.41141,
0.41241,	0.41341,	0.41441,	0.41542,	0.41642,
0.41742,	0.41842,	0.41942,	0.42042,	0.42142,
0.42242,	0.42342,	0.42442,	0.42543,	0.42643,
0.42743,	0.42843,	0.42943,	0.43043,	0.43143,
	0.43243,	0.43343,	0.43443,	0.43544,
0.43644,	0.43744,	0.43844,	0.43944,	0.44044,
0.44144,	0.44244,	0.44344,	0.44444,	0.44545,
0.44645,	0.44745,	0.44845,	0.44945,	0.45045,
0.45145,	0.45245,	0.45345,	0.45445,	0.45546,
	0.45646,	0.45746,	0.45846,	0.45946,
0.46046,	0.46146,	0.46246,	0.46346,	0.46446,
0.46547,	0.46647,	0.46747,	0.46847,	0.46947,
0.47047,	0.47147,	0.47247,	0.47347,	0.47447,
0.47548,	0.47648,	0.47748,	0.47848,	0.47948,
	0.48048,	0.48148,	0.48248,	0.48348,
0.48448,	0.48549,	0.48649,	0.48749,	0.48849,
0.48949,	0.49049,	0.49149,	0.49249,	0.49349,
0.49449,	0.4955,	0.4965,	0.4975,	0.4985,
0.4995,	0.5005,	0.5015,	0.5025,	0.5035,
	0.5045,	0.50551,	0.50651,	0.50751,
0.50851,	0.50951,	0.51051,	0.51151,	0.51251,
0.51351,	0.51451,	0.51552,	0.51652,	0.51752,
0.51852,	0.51952,	0.52052,	0.52152,	0.52252,
0.52352,	0.52452,	0.52553,	0.52653,	0.52753,
	0.52853,	0.52953,	0.53053,	0.53153,
0.53253,	0.53353,	0.53453,	0.53554,	0.53654,
0.53754,	0.53854,	0.53954,	0.54054,	0.54154,
0.54254,	0.54354,	0.54454,	0.54555,	0.54655,
0.54755,	0.54855,	0.54955,	0.55055,	0.55155,
	0.55255,	0.55355,	0.55455,	0.55556,
0.55656,	0.55756,	0.55856,	0.55956,	0.56056,
0.56156,	0.56256,	0.56356,	0.56456,	0.56557,
0.56657,	0.56757,	0.56857,	0.56957,	0.57057,
0.57157,	0.57257,	0.57357,	0.57457,	0.57558,
	0.57658,	0.57758,	0.57858,	0.57958,
0.58058,	0.58158,	0.58258,	0.58358,	0.58458,
0.58559,	0.58659,	0.58759,	0.58859,	0.58959,
0.59059,	0.59159,	0.59259,	0.59359,	0.59459,
0.5956,	0.5966,	0.5976,	0.5986,	0.5996,
	0.6006,	0.6016,	0.6026,	0.6036,
0.6046,	0.60561,	0.60661,	0.60761,	0.60861,

0.60961,	0.61061,	0.61161,	0.61261,	0.61361,
0.61461,	0.61562,	0.61662,	0.61762,	0.61862,
0.61962,	0.62062,	0.62162,	0.62262,	0.62362,
	0.62462,	0.62563,	0.62663,	0.62763,
0.62863,	0.62963,	0.63063,	0.63163,	0.63263,
0.63363,	0.63463,	0.63564,	0.63664,	0.63764,
0.63864,	0.63964,	0.64064,	0.64164,	0.64264,
0.64364,	0.64464,	0.64565,	0.64665,	0.64765,
	0.64865,	0.64965,	0.65065,	0.65165,
0.65265,	0.65365,	0.65465,	0.65566,	0.65666,
0.65766,	0.65866,	0.65966,	0.66066,	0.66166,
0.66266,	0.66366,	0.66466,	0.66567,	0.66667,
0.66767,	0.66867,	0.66967,	0.67067,	0.67167,
	0.67267,	0.67367,	0.67467,	0.67568,
0.67668,	0.67768,	0.67868,	0.67968,	0.68068,
0.68168,	0.68268,	0.68368,	0.68468,	0.68569,
0.68669,	0.68769,	0.68869,	0.68969,	0.69069,
0.69169,	0.69269,	0.69369,	0.69469,	0.6957,
	0.6967,	0.6977,	0.6987,	0.6997,
0.7007,	0.7017,	0.7027,	0.7037,	0.7047,
0.70571,	0.70671,	0.70771,	0.70871,	0.70971,
0.71071,	0.71171,	0.71271,	0.71371,	0.71471,
0.71572,	0.71672,	0.71772,	0.71872,	0.71972,
	0.72072,	0.72172,	0.72272,	0.72372,
0.72472,	0.72573,	0.72673,	0.72773,	0.72873,
0.72973,	0.73073,	0.73173,	0.73273,	0.73373,
0.73473,	0.73574,	0.73674,	0.73774,	0.73874,
0.73974,	0.74074,	0.74174,	0.74274,	0.74374,
	0.74474,	0.74575,	0.74675,	0.74775,
0.74875,	0.74975,	0.75075,	0.75175,	0.75275,
0.75375,	0.75475,	0.75576,	0.75676,	0.75776,
0.75876,	0.75976,	0.76076,	0.76176,	0.76276,
0.76376,	0.76476,	0.76577,	0.76677,	0.76777,
	0.76877,	0.76977,	0.77077,	0.77177,
0.77277,	0.77377,	0.77477,	0.77578,	0.77678,
0.77778,	0.77878,	0.77978,	0.78078,	0.78178,
0.78278,	0.78378,	0.78478,	0.78579,	0.78679,
0.78779,	0.78879,	0.78979,	0.79079,	0.79179,
	0.79279,	0.79379,	0.79479,	0.7958,
0.7968,	0.7978,	0.7988,	0.7998,	0.8008,
0.8018,	0.8028,	0.8038,	0.8048,	0.80581,
0.80681,	0.80781,	0.80881,	0.80981,	0.81081,
0.81181,	0.81281,	0.81381,	0.81481,	0.81582,
	0.81682,	0.81782,	0.81882,	0.81982,
0.82082,	0.82182,	0.82282,	0.82382,	0.82482,
0.82583,	0.82683,	0.82783,	0.82883,	0.82983,
0.83083,	0.83183,	0.83283,	0.83383,	0.83483,
0.83584,	0.83684,	0.83784,	0.83884,	0.83984,
	0.84084,	0.84184,	0.84284,	0.84384,

0.84484,	0.84585,	0.84685,	0.84785,	0.84885,
0.84985,	0.85085,	0.85185,	0.85285,	0.85385,
0.85485,	0.85586,	0.85686,	0.85786,	0.85886,
0.85986,	0.86086,	0.86186,	0.86286,	0.86386,
	0.86486,	0.86587,	0.86687,	0.86787,
0.86887,	0.86987,	0.87087,	0.87187,	0.87287,
0.87387,	0.87487,	0.87588,	0.87688,	0.87788,
0.87888,	0.87988,	0.88088,	0.88188,	0.88288,
0.88388,	0.88488,	0.88589,	0.88689,	0.88789,
	0.88889,	0.88989,	0.89089,	0.89189,
0.89289,	0.89389,	0.89489,	0.8959,	0.8969,
0.8979,	0.8989,	0.8999,	0.9009,	0.9019,
0.9029,	0.9039,	0.9049,	0.90591,	0.90691,
0.90791,	0.90891,	0.90991,	0.91091,	0.91191,
	0.91291,	0.91391,	0.91491,	0.91592,
0.91692,	0.91792,	0.91892,	0.91992,	0.92092,
0.92192,	0.92292,	0.92392,	0.92492,	0.92593,
0.92693,	0.92793,	0.92893,	0.92993,	0.93093,
0.93193,	0.93293,	0.93393,	0.93493,	0.93594,
	0.93694,	0.93794,	0.93894,	0.93994,
0.94094,	0.94194,	0.94294,	0.94394,	0.94494,
0.94595,	0.94695,	0.94795,	0.94895,	0.94995,
0.95095,	0.95195,	0.95295,	0.95395,	0.95495,
0.95596,	0.95696,	0.95796,	0.95896,	0.95996,
	0.96096,	0.96196,	0.96296,	0.96396,
0.96496,	0.96597,	0.96697,	0.96797,	0.96897,
0.96997,	0.97097,	0.97197,	0.97297,	0.97397,
0.97497,	0.97598,	0.97698,	0.97798,	0.97898,
0.97998,	0.98098,	0.98198,	0.98298,	0.98398,
	0.98498,	0.98599,	0.98699,	0.98799,
0.98899,	0.98999,	0.99099,	0.99199,	0.99299,
0.99399,	0.99499,	0.996,	0.997,	0.998,
0.999,	1]],	array([[	0.057318,	0.057318,
				0.085766,
0.10901,	0.12886,	0.1465,	0.16261,	0.17673,
0.19032,	0.20273,	0.21466,	0.22552,	0.23587,
0.24557,	0.25555,	0.26449,	0.27281,	0.28117,
0.28933,	0.29728,	0.30474,	0.31156,	0.31823,
	0.32473,	0.33145,	0.33754,	0.34361,
0.34957,	0.35523,	0.36019,	0.36556,	0.37085,
0.37584,	0.38075,	0.38533,	0.38997,	0.39413,
0.39813,	0.40298,	0.40651,	0.41085,	0.41485,
0.41856,	0.42264,	0.42612,	0.42993,	
	0.43366,	0.43715,	0.44062,	0.44447,
0.44752,	0.45034,	0.45402,	0.45738,	0.46024,
0.46343,	0.46622,	0.46867,	0.47152,	0.474,
0.47664,	0.47929,	0.48173,	0.48456,	0.48689,
0.48904,	0.49118,	0.4935,	0.49586,	
	0.49811,	0.50036,	0.5027,	0.50451,
0.50676,	0.50879,	0.51065,	0.51229,	0.51415,

0.51603,	0.51769,	0.51974,	0.52146,	0.52314,	
0.52518,	0.52674,	0.52865,	0.53034,	0.53187,	
0.53303,	0.53433,	0.53606,	0.53757,		
	0.53895,	0.54057,	0.54211,	0.5435,	
0.54491,	0.5463,	0.54752,	0.54866,	0.54972,	
0.55117,	0.55246,	0.55394,	0.55478,	0.556,	
0.55723,	0.5586,	0.55932,	0.56062,	0.56197,	
0.56287,	0.56437,	0.56542,	0.56674,		
	0.56758,	0.56885,	0.56981,	0.57086,	
0.57191,	0.57281,	0.57375,	0.57477,	0.57588,	
0.57683,	0.57771,	0.5786,	0.57912,	0.58069,	
0.58137,	0.58236,	0.58306,	0.5843,	0.58519,	
0.58611,	0.58686,	0.58788,	0.58863,		
	0.58905,	0.58995,	0.59025,	0.59097,	
0.59175,	0.59263,	0.59322,	0.59404,	0.59464,	
0.59574,	0.59636,	0.59688,	0.59741,	0.59809,	
0.59881,	0.59964,	0.60018,	0.60067,	0.60156,	
0.60212,	0.60245,	0.60303,	0.60354,		
	0.6043,	0.6051,	0.60557,	0.6059,	
0.60663,	0.60731,	0.60753,	0.60806,	0.60849,	
0.60909,	0.60989,	0.61058,	0.61107,	0.61156,	
0.61224,	0.61285,	0.61324,	0.6137,	0.61408,	
0.6145,	0.61471,	0.61522,	0.61573,		
	0.61637,	0.61672,	0.61704,	0.61745,	
0.61771,	0.6184,	0.61878,	0.61885,	0.61944,	
0.62,	0.6203,	0.6209,	0.62115,	0.62148,	0.62154,
0.62149,	0.62177,	0.62214,	0.62231,	0.62285,	
0.62318,	0.62322,	0.62355,			
	0.62398,	0.62437,	0.62464,	0.62493,	
0.62522,	0.62545,	0.62542,	0.62558,	0.62567,	
0.62604,	0.62628,	0.62623,	0.62637,	0.62666,	
0.62674,	0.62707,	0.62711,	0.62755,	0.62766,	
0.6278,	0.62801,	0.62803,	0.62814,		
	0.62851,	0.62849,	0.62853,	0.62887,	
0.62891,	0.62924,	0.62918,	0.6294,	0.62952,	
0.62943,	0.6292,	0.62921,	0.62956,	0.62933,	
0.62934,	0.62967,	0.62977,	0.62956,	0.62967,	
0.6301,	0.63038,	0.63029,	0.63034,		
	0.63052,	0.63064,	0.63083,	0.63092,	
0.63083,	0.63106,	0.63098,	0.63101,	0.63136,	
0.63137,	0.63133,	0.63124,	0.6312,	0.63133,	
0.63126,	0.63132,	0.63127,	0.63121,	0.63115,	
0.63089,	0.63097,	0.63086,	0.63068,		
	0.63045,	0.6305,	0.63033,	0.63001,	
0.62983,	0.62976,	0.62927,	0.62891,	0.62904,	
0.62911,	0.6289,	0.62894,	0.62912,	0.62906,	
0.62895,	0.62873,	0.6286,	0.62847,	0.62809,	
0.62804,	0.62774,	0.62744,	0.62703,		
	0.62695,	0.62705,	0.62695,	0.6268,	

0.62632,	0.62615,	0.62592,	0.62574,	0.62572,
0.62562,	0.62564,	0.62553,	0.6253,	0.62473,
0.62475,	0.62446,	0.62416,	0.62375,	0.62344,
0.62325,	0.62293,	0.62275,	0.62292,	
	0.62291,	0.62283,	0.62276,	0.62282,
0.62251,	0.62235,	0.62169,	0.62133,	0.62103,
0.62083,	0.6207,	0.62053,	0.62072,	0.62036,
0.62031,	0.61998,	0.61996,	0.6196,	0.61934,
0.61913,	0.61886,	0.61848,	0.61838,	
	0.61821,	0.61794,	0.61763,	0.6174,
0.61742,	0.61688,	0.61648,	0.61621,	0.61591,
0.61558,	0.6155,	0.6154,	0.61522,	0.61486,
0.61443,	0.61362,	0.61312,	0.61258,	0.61203,
0.61138,	0.61132,	0.61124,	0.6108,	
	0.6108,	0.61047,	0.6097,	0.60946,
0.60869,	0.6084,	0.60776,	0.60744,	0.60719,
0.60662,	0.6059,	0.60533,	0.60482,	0.60432,
0.60369,	0.60345,	0.60298,	0.60274,	0.60252,
0.6021,	0.60154,	0.60136,	0.60051,	
	0.59977,	0.59917,	0.59879,	0.59842,
0.59834,	0.59787,	0.59705,	0.5966,	0.59596,
0.59549,	0.5952,	0.59503,	0.59455,	0.59394,
0.5935,	0.59319,	0.59206,	0.59173,	0.59125,
0.59091,	0.59049,	0.59043,	0.58973,	
	0.58933,	0.58833,	0.5879,	0.58718,
0.58646,	0.58596,	0.58533,	0.58488,	0.58453,
0.58367,	0.5832,	0.58265,	0.58205,	0.58147,
0.58074,	0.58027,	0.57975,	0.57916,	0.5787,
0.57812,	0.57738,	0.57679,	0.57612,	
	0.57579,	0.57519,	0.57485,	0.57437,
0.57387,	0.57347,	0.57299,	0.57247,	0.57206,
0.57138,	0.57065,	0.57017,	0.56962,	0.56849,
0.56773,	0.56728,	0.56661,	0.56624,	0.56551,
0.56478,	0.56391,	0.56353,	0.56304,	
	0.5626,	0.56211,	0.56127,	0.56066,
0.55989,	0.55929,	0.55875,	0.55778,	0.55743,
0.55719,	0.55649,	0.55599,	0.55566,	0.55507,
0.55455,	0.55403,	0.55333,	0.5526,	0.55202,
0.55159,	0.55082,	0.55019,	0.54943,	
	0.54875,	0.54804,	0.54727,	0.54677,
0.54596,	0.54553,	0.54504,	0.54431,	0.54336,
0.54255,	0.54176,	0.54111,	0.54044,	0.5397,
0.53888,	0.53818,	0.53749,	0.53674,	0.53595,
0.53528,	0.53446,	0.53356,	0.53248,	
	0.53209,	0.53159,	0.53068,	0.53015,
0.52944,	0.52868,	0.52813,	0.52732,	0.52646,
0.52538,	0.52478,	0.52384,	0.5231,	0.52221,
0.52091,	0.52029,	0.5193,	0.51863,	0.5176,
0.51703,	0.51633,	0.51533,	0.51461,	

	0.51372,	0.51303,	0.5125,	0.51197,
0.51103,	0.5101,	0.50947,	0.50909,	0.508,
0.50714,	0.50635,	0.50554,	0.50436,	0.50304,
0.5022,	0.50142,	0.50043,	0.49936,	0.49854,
0.49782,	0.49718,	0.49605,	0.49514,	
	0.49437,	0.49346,	0.49212,	0.49119,
0.49035,	0.48842,	0.48741,	0.48626,	0.48521,
0.48426,	0.48329,	0.48242,	0.48132,	0.48019,
0.47903,	0.47811,	0.47674,	0.4756,	0.47469,
0.47352,	0.47225,	0.47088,	0.4701,	
	0.46865,	0.46776,	0.46672,	0.46602,
0.46466,	0.46219,	0.46121,	0.46016,	0.4589,
0.4579,	0.45702,	0.45598,	0.45515,	0.45387,
0.45249,	0.45113,	0.44968,	0.44832,	0.44689,
0.44596,	0.4447,	0.44256,	0.44147,	
	0.44021,	0.43918,	0.43831,	0.43701,
0.43595,	0.43487,	0.43378,	0.43267,	0.43119,
0.42847,	0.42759,	0.42639,	0.42525,	0.42396,
0.42246,	0.42115,	0.4201,	0.41879,	0.41794,
0.41665,	0.41537,	0.41426,	0.41186,	
	0.41041,	0.40884,	0.40764,	0.40651,
0.40561,	0.40399,	0.40276,	0.40172,	0.40017,
0.39891,	0.39737,	0.39546,	0.39441,	0.39308,
0.39184,	0.39092,	0.38949,	0.38835,	0.38712,
0.38438,	0.38339,	0.38234,	0.38093,	
	0.37965,	0.3784,	0.37735,	0.37603,
0.37491,	0.37262,	0.37098,	0.36956,	0.3684,
0.36715,	0.3657,	0.36376,	0.3626,	0.36099,
0.35821,	0.35659,	0.35548,	0.35414,	0.35148,
0.35026,	0.34871,	0.34799,	0.34626,	
	0.34546,	0.34329,	0.34301,	0.34018,
0.33829,	0.33764,	0.33551,	0.33481,	0.33327,
0.33238,	0.32952,	0.32757,	0.32664,	0.32392,
0.32226,	0.3213,	0.31917,	0.31822,	0.31644,
0.31504,	0.31127,	0.30898,	0.30783,	
	0.30559,	0.30449,	0.30181,	0.2989,
0.29736,	0.29433,	0.29308,	0.29081,	0.2871,
0.28568,	0.28146,	0.27864,	0.27754,	0.27546,
0.27464,	0.27217,	0.2696,	0.2686,	0.2668,
0.26562,	0.26267,	0.26068,	0.26015,	
	0.25872,	0.25762,	0.2553,	0.25383,
0.25303,	0.25127,	0.24928,	0.24849,	0.24629,
0.24524,	0.24458,	0.243,	0.24115,	0.24056,
0.23899,	0.23676,	0.23596,	0.23444,	0.23336,
0.23112,	0.22911,	0.22815,	0.22693,	
	0.22477,	0.22373,	0.22166,	0.21879,
0.21624,	0.21541,	0.21378,	0.21233,	0.20917,
0.20741,	0.20684,	0.20361,	0.20185,	0.20061,
0.19765,	0.19556,	0.19464,	0.19197,	0.1906,

0.18974,	0.18678,	0.18499,	0.18291,		
	0.18214,	0.1794,	0.17731,	0.17429,	
0.17304,	0.17047,	0.16888,	0.16809,	0.16498,	
0.163,	0.16035,	0.15984,	0.15894,	0.15686,	
0.15582,	0.15359,	0.15208,	0.15124,	0.14944,	
0.14743,	0.14491,	0.14263,	0.14168,		
	0.13876,	0.13702,	0.13395,	0.13294,	
0.12991,	0.12777,	0.12473,	0.12417,	0.12151,	
0.11943,	0.11724,	0.11676,	0.11449,	0.11226,	
0.10929,	0.10636,	0.10562,	0.10245,	0.10043,	
0.097154,	0.095989,	0.093163,	0.092236,		
	0.089712,	0.088056,	0.087072,	0.084738,	
0.084257,	0.082343,	0.080472,	0.07898,	0.077153,	
0.07577,	0.075248,	0.073841,	0.072391,	0.071155,	
0.069386,	0.067191,	0.065984,	0.064136,	0.061639,	
0.060251,	0.057915,	0.054968,	0.051741,		
	0.049701,	0.046806,	0.044408,	0.042157,	
0.041338,	0.038939,	0.036959,	0.035563,	0.034532,	
0.033387,	0.032906,	0.032391,	0.031211,	0.030028,	
0.029361,	0.028472,	0.027655,	0.0267,	0.026093,	
0.025274,	0.023935,	0.02304,	0.022219,		
	0.021471,	0.020723,	0.020049,	0.019747,	
0.019221,	0.018247,	0.017571,	0.016969,	0.015993,	
0.015016,	0.014413,	0.013884,	0.013281,	0.012376,	
0.012222,	0.011843,	0.011237,	0.010256,	0.0098001,	
0.0094779,	0.0091147,	0.0083571,	0.0075989,		
	0.0069912,	0.0065342,	0.006077,	0.0058424,	
0.0056158,	0.0048556,	0.004549,	0.0042424,	0.0040547,	
0.0034773,	0.0032175,	0.0030149,	0.0027208,	0.0025523,	
0.0023757,	0.0022979,	0.0020305,	0.0018542,	0.0016851,	
0.0014174,	0.0013113,	0.0012238,	0.00088004,		
	0.00076815,	0.00068627,	0.00049427,	0.00045275,	
0.00037637,	0.00036929,	0.0003622,	0.00035512,	0.00034804,	
0.00034095,	0.00033387,	0.00032679,	0.0003197,	0.00031262,	
0.00030553,	0.00029024,	0.00027292,	0.0002556,	0.00023828,	
0.0001638,	0.00014007,	0.00012708,	0.00011409,		
	0.0001011,	8.8107e-05,	0,	0,	
0,	0,	0,	0,	0,	0,
0,	0,	0,	0,	0,	0,
0,	0,	0,	0,	0,	0,
0,					
	0,	0,	0,	0,	
0,	0,	0,	0,	0,	0,
0,	0,	0,	0,	0,	0,
0,	0,	0,	0,	0,	0,
0,					
	0,	0,	0,	0,	
0,	0,	0,	0,	0,	0,
0,	0,	0,	0,	0,	0,

0,	0,	0,	0,	0,	0,
0,					
	0,	0,	0,	0,	0,
0,	0,	0,	0,	0,	0,
0]]), 'Confidence', 'F1'], [array([	0,	0.001001,			
0.002002,	0.003003,	0.004004,	0.005005,	0.006006,	
0.007007,	0.008008,	0.009009,	0.01001,	0.011011,	
0.012012,	0.013013,	0.014014,	0.015015,	0.016016,	
0.017017,	0.018018,	0.019019,	0.02002,	0.021021,	
0.022022,	0.023023,				
	0.024024,	0.025025,	0.026026,	0.027027,	
0.028028,	0.029029,	0.03003,	0.031031,	0.032032,	
0.033033,	0.034034,	0.035035,	0.036036,	0.037037,	
0.038038,	0.039039,	0.04004,	0.041041,	0.042042,	
0.043043,	0.044044,	0.045045,	0.046046,	0.047047,	
	0.048048,	0.049049,	0.05005,	0.051051,	
0.052052,	0.053053,	0.054054,	0.055055,	0.056056,	
0.057057,	0.058058,	0.059059,	0.06006,	0.061061,	
0.062062,	0.063063,	0.064064,	0.065065,	0.066066,	
0.067067,	0.068068,	0.069069,	0.07007,	0.071071,	
	0.072072,	0.073073,	0.074074,	0.075075,	
0.076076,	0.077077,	0.078078,	0.079079,	0.08008,	
0.081081,	0.082082,	0.083083,	0.084084,	0.085085,	
0.086086,	0.087087,	0.088088,	0.089089,	0.09009,	
0.091091,	0.092092,	0.093093,	0.094094,	0.095095,	
	0.096096,	0.097097,	0.098098,	0.099099,	
0.1001,	0.1011,	0.1021,	0.1031,	0.1041,	
0.10511,	0.10611,	0.10711,	0.10811,	0.10911,	
0.11011,	0.11111,	0.11211,	0.11311,	0.11411,	
0.11512,	0.11612,	0.11712,	0.11812,	0.11912,	
	0.12012,	0.12112,	0.12212,	0.12312,	
0.12412,	0.12513,	0.12613,	0.12713,	0.12813,	
0.12913,	0.13013,	0.13113,	0.13213,	0.13313,	
0.13413,	0.13514,	0.13614,	0.13714,	0.13814,	
0.13914,	0.14014,	0.14114,	0.14214,	0.14314,	
	0.14414,	0.14515,	0.14615,	0.14715,	
0.14815,	0.14915,	0.15015,	0.15115,	0.15215,	
0.15315,	0.15415,	0.15516,	0.15616,	0.15716,	
0.15816,	0.15916,	0.16016,	0.16116,	0.16216,	
0.16316,	0.16416,	0.16517,	0.16617,	0.16717,	
	0.16817,	0.16917,	0.17017,	0.17117,	
0.17217,	0.17317,	0.17417,	0.17518,	0.17618,	
0.17718,	0.17818,	0.17918,	0.18018,	0.18118,	
0.18218,	0.18318,	0.18418,	0.18519,	0.18619,	
0.18719,	0.18819,	0.18919,	0.19019,	0.19119,	
	0.19219,	0.19319,	0.19419,	0.1952,	
0.1962,	0.1972,	0.1982,	0.1992,	0.2002,	
0.2012,	0.2022,	0.2032,	0.2042,	0.20521,	
0.20621,	0.20721,	0.20821,	0.20921,	0.21021,	

0.21121,	0.21221,	0.21321,	0.21421,	0.21522,
	0.21622,	0.21722,	0.21822,	0.21922,
0.22022,	0.22122,	0.22222,	0.22322,	0.22422,
0.22523,	0.22623,	0.22723,	0.22823,	0.22923,
0.23023,	0.23123,	0.23223,	0.23323,	0.23423,
0.23524,	0.23624,	0.23724,	0.23824,	0.23924,
	0.24024,	0.24124,	0.24224,	0.24324,
0.24424,	0.24525,	0.24625,	0.24725,	0.24825,
0.24925,	0.25025,	0.25125,	0.25225,	0.25325,
0.25425,	0.25526,	0.25626,	0.25726,	0.25826,
0.25926,	0.26026,	0.26126,	0.26226,	0.26326,
	0.26426,	0.26527,	0.26627,	0.26727,
0.26827,	0.26927,	0.27027,	0.27127,	0.27227,
0.27327,	0.27427,	0.27528,	0.27628,	0.27728,
0.27828,	0.27928,	0.28028,	0.28128,	0.28228,
0.28328,	0.28428,	0.28529,	0.28629,	0.28729,
	0.28829,	0.28929,	0.29029,	0.29129,
0.29229,	0.29329,	0.29429,	0.2953,	0.2963,
0.2973,	0.2983,	0.2993,	0.3003,	0.3013,
0.3023,	0.3033,	0.3043,	0.30531,	0.30631,
0.30731,	0.30831,	0.30931,	0.31031,	0.31131,
	0.31231,	0.31331,	0.31431,	0.31532,
0.31632,	0.31732,	0.31832,	0.31932,	0.32032,
0.32132,	0.32232,	0.32332,	0.32432,	0.32533,
0.32633,	0.32733,	0.32833,	0.32933,	0.33033,
0.33133,	0.33233,	0.33333,	0.33433,	0.33534,
	0.33634,	0.33734,	0.33834,	0.33934,
0.34034,	0.34134,	0.34234,	0.34334,	0.34434,
0.34535,	0.34635,	0.34735,	0.34835,	0.34935,
0.35035,	0.35135,	0.35235,	0.35335,	0.35435,
0.35536,	0.35636,	0.35736,	0.35836,	0.35936,
	0.36036,	0.36136,	0.36236,	0.36336,
0.36436,	0.36537,	0.36637,	0.36737,	0.36837,
0.36937,	0.37037,	0.37137,	0.37237,	0.37337,
0.37437,	0.37538,	0.37638,	0.37738,	0.37838,
0.37938,	0.38038,	0.38138,	0.38238,	0.38338,
	0.38438,	0.38539,	0.38639,	0.38739,
0.38839,	0.38939,	0.39039,	0.39139,	0.39239,
0.39339,	0.39439,	0.3954,	0.3964,	0.3974,
0.3984,	0.3994,	0.4004,	0.4014,	0.4024,
0.4034,	0.4044,	0.40541,	0.40641,	0.40741,
	0.40841,	0.40941,	0.41041,	0.41141,
0.41241,	0.41341,	0.41441,	0.41542,	0.41642,
0.41742,	0.41842,	0.41942,	0.42042,	0.42142,
0.42242,	0.42342,	0.42442,	0.42543,	0.42643,
0.42743,	0.42843,	0.42943,	0.43043,	0.43143,
	0.43243,	0.43343,	0.43443,	0.43544,
0.43644,	0.43744,	0.43844,	0.43944,	0.44044,
0.44144,	0.44244,	0.44344,	0.44444,	0.44545,

0.44645,	0.44745,	0.44845,	0.44945,	0.45045,
0.45145,	0.45245,	0.45345,	0.45445,	0.45546,
	0.45646,	0.45746,	0.45846,	0.45946,
0.46046,	0.46146,	0.46246,	0.46346,	0.46446,
0.46547,	0.46647,	0.46747,	0.46847,	0.46947,
0.47047,	0.47147,	0.47247,	0.47347,	0.47447,
0.47548,	0.47648,	0.47748,	0.47848,	0.47948,
	0.48048,	0.48148,	0.48248,	0.48348,
0.48448,	0.48549,	0.48649,	0.48749,	0.48849,
0.48949,	0.49049,	0.49149,	0.49249,	0.49349,
0.49449,	0.4955,	0.4965,	0.4975,	0.4985,
0.4995,	0.5005,	0.5015,	0.5025,	0.5035,
	0.5045,	0.50551,	0.50651,	0.50751,
0.50851,	0.50951,	0.51051,	0.51151,	0.51251,
0.51351,	0.51451,	0.51552,	0.51652,	0.51752,
0.51852,	0.51952,	0.52052,	0.52152,	0.52252,
0.52352,	0.52452,	0.52553,	0.52653,	0.52753,
	0.52853,	0.52953,	0.53053,	0.53153,
0.53253,	0.53353,	0.53453,	0.53554,	0.53654,
0.53754,	0.53854,	0.53954,	0.54054,	0.54154,
0.54254,	0.54354,	0.54454,	0.54555,	0.54655,
0.54755,	0.54855,	0.54955,	0.55055,	0.55155,
	0.55255,	0.55355,	0.55455,	0.55556,
0.55656,	0.55756,	0.55856,	0.55956,	0.56056,
0.56156,	0.56256,	0.56356,	0.56456,	0.56557,
0.56657,	0.56757,	0.56857,	0.56957,	0.57057,
0.57157,	0.57257,	0.57357,	0.57457,	0.57558,
	0.57658,	0.57758,	0.57858,	0.57958,
0.58058,	0.58158,	0.58258,	0.58358,	0.58458,
0.58559,	0.58659,	0.58759,	0.58859,	0.58959,
0.59059,	0.59159,	0.59259,	0.59359,	0.59459,
0.5956,	0.5966,	0.5976,	0.5986,	0.5996,
	0.6006,	0.6016,	0.6026,	0.6036,
0.6046,	0.60561,	0.60661,	0.60761,	0.60861,
0.60961,	0.61061,	0.61161,	0.61261,	0.61361,
0.61461,	0.61562,	0.61662,	0.61762,	0.61862,
0.61962,	0.62062,	0.62162,	0.62262,	0.62362,
	0.62462,	0.62563,	0.62663,	0.62763,
0.62863,	0.62963,	0.63063,	0.63163,	0.63263,
0.63363,	0.63463,	0.63564,	0.63664,	0.63764,
0.63864,	0.63964,	0.64064,	0.64164,	0.64264,
0.64364,	0.64464,	0.64565,	0.64665,	0.64765,
	0.64865,	0.64965,	0.65065,	0.65165,
0.65265,	0.65365,	0.65465,	0.65566,	0.65666,
0.65766,	0.65866,	0.65966,	0.66066,	0.66166,
0.66266,	0.66366,	0.66466,	0.66567,	0.66667,
0.66767,	0.66867,	0.66967,	0.67067,	0.67167,
	0.67267,	0.67367,	0.67467,	0.67568,
0.67668,	0.67768,	0.67868,	0.67968,	0.68068,

0.68168,	0.68268,	0.68368,	0.68468,	0.68569,
0.68669,	0.68769,	0.68869,	0.68969,	0.69069,
0.69169,	0.69269,	0.69369,	0.69469,	0.6957,
	0.6967,	0.6977,	0.6987,	0.6997,
0.7007,	0.7017,	0.7027,	0.7037,	0.7047,
0.70571,	0.70671,	0.70771,	0.70871,	0.70971,
0.71071,	0.71171,	0.71271,	0.71371,	0.71471,
0.71572,	0.71672,	0.71772,	0.71872,	0.71972,
	0.72072,	0.72172,	0.72272,	0.72372,
0.72472,	0.72573,	0.72673,	0.72773,	0.72873,
0.72973,	0.73073,	0.73173,	0.73273,	0.73373,
0.73473,	0.73574,	0.73674,	0.73774,	0.73874,
0.73974,	0.74074,	0.74174,	0.74274,	0.74374,
	0.74474,	0.74575,	0.74675,	0.74775,
0.74875,	0.74975,	0.75075,	0.75175,	0.75275,
0.75375,	0.75475,	0.75576,	0.75676,	0.75776,
0.75876,	0.75976,	0.76076,	0.76176,	0.76276,
0.76376,	0.76476,	0.76577,	0.76677,	0.76777,
	0.76877,	0.76977,	0.77077,	0.77177,
0.77277,	0.77377,	0.77477,	0.77578,	0.77678,
0.77778,	0.77878,	0.77978,	0.78078,	0.78178,
0.78278,	0.78378,	0.78478,	0.78579,	0.78679,
0.78779,	0.78879,	0.78979,	0.79079,	0.79179,
	0.79279,	0.79379,	0.79479,	0.7958,
0.7968,	0.7978,	0.7988,	0.7998,	0.8008,
0.8018,	0.8028,	0.8038,	0.8048,	0.80581,
0.80681,	0.80781,	0.80881,	0.80981,	0.81081,
0.81181,	0.81281,	0.81381,	0.81481,	0.81582,
	0.81682,	0.81782,	0.81882,	0.81982,
0.82082,	0.82182,	0.82282,	0.82382,	0.82482,
0.82583,	0.82683,	0.82783,	0.82883,	0.82983,
0.83083,	0.83183,	0.83283,	0.83383,	0.83483,
0.83584,	0.83684,	0.83784,	0.83884,	0.83984,
	0.84084,	0.84184,	0.84284,	0.84384,
0.84484,	0.84585,	0.84685,	0.84785,	0.84885,
0.84985,	0.85085,	0.85185,	0.85285,	0.85385,
0.85485,	0.85586,	0.85686,	0.85786,	0.85886,
0.85986,	0.86086,	0.86186,	0.86286,	0.86386,
	0.86486,	0.86587,	0.86687,	0.86787,
0.86887,	0.86987,	0.87087,	0.87187,	0.87287,
0.87387,	0.87487,	0.87588,	0.87688,	0.87788,
0.87888,	0.87988,	0.88088,	0.88188,	0.88288,
0.88388,	0.88488,	0.88589,	0.88689,	0.88789,
	0.88889,	0.88989,	0.89089,	0.89189,
0.89289,	0.89389,	0.89489,	0.8959,	0.8969,
0.8979,	0.8989,	0.8999,	0.9009,	0.9019,
0.9029,	0.9039,	0.9049,	0.90591,	0.90691,
0.90791,	0.90891,	0.90991,	0.91091,	0.91191,
	0.91291,	0.91391,	0.91491,	0.91592,
0.91692,	0.91792,	0.91892,	0.91992,	0.92092,

0.92192,	0.92292,	0.92392,	0.92492,	0.92593,
0.92693,	0.92793,	0.92893,	0.92993,	0.93093,
0.93193,	0.93293,	0.93393,	0.93493,	0.93594,
	0.93694,	0.93794,	0.93894,	0.93994,
0.94094,	0.94194,	0.94294,	0.94394,	0.94494,
0.94595,	0.94695,	0.94795,	0.94895,	0.94995,
0.95095,	0.95195,	0.95295,	0.95395,	0.95495,
0.95596,	0.95696,	0.95796,	0.95896,	0.95996,
	0.96096,	0.96196,	0.96296,	0.96396,
0.96496,	0.96597,	0.96697,	0.96797,	0.96897,
0.96997,	0.97097,	0.97197,	0.97297,	0.97397,
0.97497,	0.97598,	0.97698,	0.97798,	0.97898,
0.97998,	0.98098,	0.98198,	0.98298,	0.98398,
	0.98498,	0.98599,	0.98699,	0.98799,
0.98899,	0.98999,	0.99099,	0.99199,	0.99299,
0.99399,	0.99499,	0.996,	0.997,	0.998,
0.999,	1]],	array([[	0.029566,	0.029566,
0.057951,	0.069341,	0.079712,	0.089397,	0.098061,
0.10656,	0.11448,	0.12222,	0.12937,	0.1363,
0.1429,	0.14978,	0.15604,	0.16196,	0.16799,
0.17393,	0.17982,	0.18541,	0.1906,	0.19576,
	0.20084,	0.20613,	0.21094,	0.21584,
0.22068,	0.22533,	0.22946,	0.23396,	0.23845,
0.24272,	0.24701,	0.25108,	0.25521,	0.25896,
0.26259,	0.26696,	0.27026,	0.27426,	0.27807,
0.28163,	0.28556,	0.28891,	0.2926,	
	0.29625,	0.29962,	0.30314,	0.30697,
0.3101,	0.31307,	0.31687,	0.32027,	0.32326,
0.32657,	0.32963,	0.33222,	0.33539,	0.33812,
0.34098,	0.34401,	0.34676,	0.34993,	0.3526,
0.35513,	0.35761,	0.36031,	0.3632,	
	0.36582,	0.36848,	0.37129,	0.37349,
0.37625,	0.37883,	0.38126,	0.38328,	0.38571,
0.38817,	0.39028,	0.393,	0.39525,	0.39742,
0.40014,	0.40224,	0.40487,	0.40711,	0.40923,
0.41106,	0.41294,	0.41546,	0.41756,	
	0.41956,	0.42188,	0.4241,	0.42609,
0.42812,	0.43025,	0.43218,	0.43379,	0.43565,
0.43785,	0.4398,	0.44209,	0.44338,	0.44541,
0.4474,	0.44951,	0.45081,	0.45287,	0.45511,
0.45668,	0.45899,	0.46064,	0.46279,	
	0.46433,	0.46648,	0.46808,	0.46997,
0.4716,	0.4733,	0.47492,	0.47662,	0.47873,
0.48037,	0.48233,	0.484,	0.48523,	0.48791,
0.48934,	0.49099,	0.49251,	0.49493,	0.49668,
0.49841,	0.49987,	0.50198,	0.50352,	
	0.50459,	0.50631,	0.50736,	0.50887,
0.51052,	0.51216,	0.51348,	0.51549,	0.51677,
0.51884,	0.52012,	0.52138,	0.52266,	0.52419,

0.5258,	0.52739,	0.52878,	0.53013,	0.53179,
0.5331,	0.53391,	0.53541,	0.53675,	
	0.53835,	0.53992,	0.54141,	0.54262,
0.5442,	0.54599,	0.54662,	0.5481,	0.54958,
0.55115,	0.55298,	0.55468,	0.55577,	0.55707,
0.55887,	0.56044,	0.56136,	0.56291,	0.56426,
0.56578,	0.56663,	0.56819,	0.56965,	
	0.57079,	0.5722,	0.5735,	0.57496,
0.57568,	0.57751,	0.57895,	0.57957,	0.58139,
0.58291,	0.58368,	0.58525,	0.58662,	0.58747,
0.58861,	0.58923,	0.59041,	0.59207,	0.59277,
0.59439,	0.59533,	0.59664,	0.59747,	
	0.59892,	0.60024,	0.60114,	0.60271,
0.6035,	0.60502,	0.60553,	0.60694,	0.60752,
0.60909,	0.60999,	0.61121,	0.61189,	0.61345,
0.6141,	0.61581,	0.6164,	0.61803,	0.61878,
0.62031,	0.62105,	0.62217,	0.62313,	
	0.62468,	0.62538,	0.62601,	0.62777,
0.62843,	0.6302,	0.63079,	0.63236,	0.63314,
0.63374,	0.63496,	0.63566,	0.63745,	0.63798,
0.63875,	0.64062,	0.6414,	0.64182,	0.64326,
0.64447,	0.64618,	0.64684,	0.64767,	
	0.64928,	0.65013,	0.65117,	0.6525,
0.65309,	0.6541,	0.65585,	0.65669,	0.65853,
0.65922,	0.65995,	0.66057,	0.66216,	0.66308,
0.66381,	0.66524,	0.66596,	0.66673,	0.66817,
0.66886,	0.66966,	0.67024,	0.6718,	
	0.67251,	0.6734,	0.67394,	0.67503,
0.67555,	0.67602,	0.67674,	0.67727,	0.67817,
0.67896,	0.68052,	0.68138,	0.68227,	0.68306,
0.68442,	0.68518,	0.68575,	0.68632,	0.68748,
0.68801,	0.68852,	0.68896,	0.68933,	
	0.69073,	0.69153,	0.69223,	0.69287,
0.69409,	0.69474,	0.69547,	0.69592,	0.6968,
0.6984,	0.69928,	0.69991,	0.70045,	0.70136,
0.70213,	0.7026,	0.70307,	0.70365,	0.70441,
0.70489,	0.70628,	0.70722,	0.7085,	
	0.7093,	0.71013,	0.71203,	0.71289,
0.71331,	0.7138,	0.71398,	0.71468,	0.71608,
0.71667,	0.71723,	0.71769,	0.71882,	0.71947,
0.72026,	0.72213,	0.72273,	0.7233,	0.7239,
0.72487,	0.7251,	0.72646,	0.72751,	
	0.72811,	0.72874,	0.72904,	0.72966,
0.73076,	0.7318,	0.73219,	0.73301,	0.73353,
0.73389,	0.7347,	0.73519,	0.73582,	0.73631,
0.73658,	0.7376,	0.7382,	0.73853,	0.73898,
0.73934,	0.7404,	0.74099,	0.74164,	
	0.74273,	0.74341,	0.74424,	0.74506,
0.74531,	0.746,	0.74623,	0.74697,	0.74735,

0.74816,	0.74859,	0.74909,	0.75015,	0.75066,
0.75089,	0.75163,	0.75233,	0.75276,	0.75349,
0.75449,	0.75506,	0.75617,	0.75735,	
	0.75763,	0.75821,	0.75896,	0.75976,
0.76088,	0.76139,	0.7616,	0.762,	0.76259,
0.76308,	0.76398,	0.76465,	0.76533,	0.76574,
0.76647,	0.76727,	0.76849,	0.7693,	0.76966,
0.77025,	0.77108,	0.77197,	0.77249,	
	0.77311,	0.77341,	0.77418,	0.77443,
0.77494,	0.77546,	0.77597,	0.77637,	0.7767,
0.7786,	0.77915,	0.77984,	0.78016,	0.78076,
0.78149,	0.7824,	0.78304,	0.78342,	0.78418,
0.78485,	0.78537,	0.78606,	0.78682,	
	0.78816,	0.78866,	0.78936,	0.79007,
0.79066,	0.79125,	0.79225,	0.79275,	0.79355,
0.79395,	0.79438,	0.79497,	0.79572,	0.79683,
0.79747,	0.79803,	0.79851,	0.79907,	0.79959,
0.80013,	0.8005,	0.80108,	0.80162,	
	0.80265,	0.8032,	0.80359,	0.80405,
0.80448,	0.80482,	0.80523,	0.8055,	0.80651,
0.80728,	0.80775,	0.80845,	0.80897,	0.80954,
0.81037,	0.81114,	0.81169,	0.81196,	0.81272,
0.81332,	0.8143,	0.81506,	0.81583,	
	0.81609,	0.81666,	0.81699,	0.81793,
0.81837,	0.81918,	0.8197,	0.82027,	0.8204,
0.8206,	0.82124,	0.82187,	0.82264,	0.82329,
0.82394,	0.8244,	0.82513,	0.82524,	0.82552,
0.82582,	0.82637,	0.82686,	0.8272,	
	0.82755,	0.82835,	0.82874,	0.82923,
0.82972,	0.83001,	0.83058,	0.8311,	0.83161,
0.83192,	0.83265,	0.83277,	0.8334,	0.8336,
0.83449,	0.83479,	0.83548,	0.8363,	0.83666,
0.83746,	0.8378,	0.83831,	0.83908,	
	0.84003,	0.84068,	0.84137,	0.84195,
0.84223,	0.8426,	0.84342,	0.84461,	0.84514,
0.84534,	0.84591,	0.84641,	0.8465,	0.84647,
0.84704,	0.84735,	0.84816,	0.84852,	0.84882,
0.84958,	0.85071,	0.85121,	0.8515,	
	0.85213,	0.85316,	0.85318,	0.85362,
0.8544,	0.85517,	0.85549,	0.85611,	0.85685,
0.85755,	0.85798,	0.8584,	0.85886,	0.85945,
0.85962,	0.86066,	0.86072,	0.86108,	0.86168,
0.86211,	0.86277,	0.86273,	0.86314,	
	0.86374,	0.86429,	0.86464,	0.86543,
0.86558,	0.86684,	0.86748,	0.86815,	0.86874,
0.86895,	0.86919,	0.86912,	0.86912,	0.86951,
0.86942,	0.86995,	0.86996,	0.87043,	0.87114,
0.87176,	0.87239,	0.87366,	0.87378,	
	0.87423,	0.87471,	0.87543,	0.87598,

0.87695,	0.87743,	0.87822,	0.87929,	0.87951,
0.88096,	0.88169,	0.88223,	0.88259,	0.8828,
0.88357,	0.8841,	0.88517,	0.88588,	0.88647,
0.88692,	0.88738,	0.88747,	0.88847,	
	0.88835,	0.88842,	0.88904,	0.88958,
0.89038,	0.89082,	0.89124,	0.89183,	0.89181,
0.89212,	0.8928,	0.89373,	0.89435,	0.89442,
0.89511,	0.89566,	0.89606,	0.89632,	0.89668,
0.89696,	0.8974,	0.89831,	0.8995,	
	0.90025,	0.90114,	0.9023,	0.90234,
0.90266,	0.90302,	0.90336,	0.90369,	0.90375,
0.9048,	0.90469,	0.90533,	0.90539,	0.90601,
0.90698,	0.90727,	0.9073,	0.90774,	0.90881,
0.90973,	0.91029,	0.91065,	0.91103,	
	0.91124,	0.9119,	0.91274,	0.91283,
0.91394,	0.91456,	0.91513,	0.91553,	0.91679,
0.91794,	0.91881,	0.91916,	0.91986,	0.92057,
0.92126,	0.92166,	0.9216,	0.92224,	0.92266,
0.92323,	0.92449,	0.92491,	0.92547,	
	0.92646,	0.92758,	0.92741,	0.93045,
0.93079,	0.93188,	0.93306,	0.93306,	0.93347,
0.9333,	0.93379,	0.93346,	0.93358,	0.93324,
0.93363,	0.93483,	0.93458,	0.93494,	0.93489,
0.93501,	0.93486,	0.93526,	0.93571,	
	0.93686,	0.93768,	0.93806,	0.93803,
0.93805,	0.93827,	0.93936,	0.93961,	0.93997,
0.9397,	0.93976,	0.94078,	0.94129,	0.94158,
0.9419,	0.94206,	0.94285,	0.94295,	0.94312,
0.94309,	0.94307,	0.94304,	0.94353,	
	0.94506,	0.94506,	0.94505,	0.94565,
0.94719,	0.94752,	0.94794,	0.94859,	0.94991,
0.94975,	0.9496,	0.95025,	0.9507,	0.95037,
0.95021,	0.9509,	0.95129,	0.95154,	0.95182,
0.95256,	0.95304,	0.95329,	0.95341,	
	0.95389,	0.9542,	0.95433,	0.9553,
0.9554,	0.95684,	0.95717,	0.95758,	0.95841,
0.95867,	0.95995,	0.96142,	0.962,	0.96229,
0.96327,	0.96356,	0.96318,	0.96382,	0.96468,
0.96504,	0.96623,	0.96659,	0.96635,	
	0.96609,	0.96613,	0.96581,	0.96554,
0.96471,	0.96462,	0.96426,	0.96409,	0.96329,
0.96429,	0.9653,	0.9663,	0.96677,	0.96848,
0.9676,	0.96697,	0.96772,	0.96736,	0.96803,
0.96972,	0.97147,	0.97132,	0.97102,	
	0.97324,	0.97273,	0.97321,	0.97362,
0.97536,	0.97667,	0.97612,	0.97654,	0.97598,
0.97553,	0.97536,	0.97677,	0.97726,	0.97686,
0.97727,	0.97861,	0.98035,	0.98198,	0.98239,
0.98198,	0.98247,	0.98152,	0.98174,	

	0.98098,	0.98285,	0.98354,	0.99125,	
0.99107,	0.99052,	0.99001,	0.98961,	0.9893,	
0.99112,	0.9931,	0.99312,	0.99523,	0.99751,	
0.99745,	1,	1,	1,	1,	
1,	1,	1,	1,		
	1,	1,	1,	1,	
1,	1,	1,	1,	1,	1,
1,	1,	1,	1,	1,	1,
1,	1,	1,	1,	1,	1,
1,					
	1,	1,	1,	1,	
1,	1,	1,	1,	1,	1,
1,	1,	1,	1,	1,	1,
1,	1,	1,	1,	1,	1,
1,					
	1,	1,	1,	1,	
1,	1,	1,	1,	1,	1,
1,	1,	1,	1,	1,	1,
1,	1,	1,	1,	1,	1,
1,					
	1,	1,	1,	1,	
1,	1,	1,	1,	1,	1,
1,	1,	1,	1,	1,	1,
1,	1,	1,	1,	1,	1,
1,					
	1,	1,	1,	1,	
1,	1,	1,	1,	1,	1,
1,	1,	1,	1,	1,	1,
1,	1,	1,	1,	1,	1,
1,					
	1,	1,	1,	1,	
1,	1,	1,	1,	1,	1,
1]]), 'Confidence', 'Precision'], [array([				0,	0.001001,
0.002002,	0.003003,	0.004004,	0.005005,	0.006006,	
0.007007,	0.008008,	0.009009,	0.01001,	0.011011,	
0.012012,	0.013013,	0.014014,	0.015015,	0.016016,	
0.017017,	0.018018,	0.019019,	0.02002,	0.021021,	
0.022022,	0.023023,				
	0.024024,	0.025025,	0.026026,	0.027027,	
0.028028,	0.029029,	0.03003,	0.031031,	0.032032,	
0.033033,	0.034034,	0.035035,	0.036036,	0.037037,	
0.038038,	0.039039,	0.04004,	0.041041,	0.042042,	
0.043043,	0.044044,	0.045045,	0.046046,	0.047047,	
	0.048048,	0.049049,	0.05005,	0.051051,	

0.052052,	0.053053,	0.054054,	0.055055,	0.056056,
0.057057,	0.058058,	0.059059,	0.06006,	0.061061,
0.062062,	0.063063,	0.064064,	0.065065,	0.066066,
0.067067,	0.068068,	0.069069,	0.07007,	0.071071,
	0.072072,	0.073073,	0.074074,	0.075075,
0.076076,	0.077077,	0.078078,	0.079079,	0.08008,
0.081081,	0.082082,	0.083083,	0.084084,	0.085085,
0.086086,	0.087087,	0.088088,	0.089089,	0.09009,
0.091091,	0.092092,	0.093093,	0.094094,	0.095095,
	0.096096,	0.097097,	0.098098,	0.099099,
0.1001,	0.1011,	0.1021,	0.1031,	0.1041,
0.10511,	0.10611,	0.10711,	0.10811,	0.10911,
0.11011,	0.11111,	0.11211,	0.11311,	0.11411,
0.11512,	0.11612,	0.11712,	0.11812,	0.11912,
	0.12012,	0.12112,	0.12212,	0.12312,
0.12412,	0.12513,	0.12613,	0.12713,	0.12813,
0.12913,	0.13013,	0.13113,	0.13213,	0.13313,
0.13413,	0.13514,	0.13614,	0.13714,	0.13814,
0.13914,	0.14014,	0.14114,	0.14214,	0.14314,
	0.14414,	0.14515,	0.14615,	0.14715,
0.14815,	0.14915,	0.15015,	0.15115,	0.15215,
0.15315,	0.15415,	0.15516,	0.15616,	0.15716,
0.15816,	0.15916,	0.16016,	0.16116,	0.16216,
0.16316,	0.16416,	0.16517,	0.16617,	0.16717,
	0.16817,	0.16917,	0.17017,	0.17117,
0.17217,	0.17317,	0.17417,	0.17518,	0.17618,
0.17718,	0.17818,	0.17918,	0.18018,	0.18118,
0.18218,	0.18318,	0.18418,	0.18519,	0.18619,
0.18719,	0.18819,	0.18919,	0.19019,	0.19119,
	0.19219,	0.19319,	0.19419,	0.1952,
0.1962,	0.1972,	0.1982,	0.1992,	0.2002,
0.2012,	0.2022,	0.2032,	0.2042,	0.20521,
0.20621,	0.20721,	0.20821,	0.20921,	0.21021,
0.21121,	0.21221,	0.21321,	0.21421,	0.21522,
	0.21622,	0.21722,	0.21822,	0.21922,
0.22022,	0.22122,	0.22222,	0.22322,	0.22422,
0.22523,	0.22623,	0.22723,	0.22823,	0.22923,
0.23023,	0.23123,	0.23223,	0.23323,	0.23423,
0.23524,	0.23624,	0.23724,	0.23824,	0.23924,
	0.24024,	0.24124,	0.24224,	0.24324,
0.24424,	0.24525,	0.24625,	0.24725,	0.24825,
0.24925,	0.25025,	0.25125,	0.25225,	0.25325,
0.25425,	0.25526,	0.25626,	0.25726,	0.25826,
0.25926,	0.26026,	0.26126,	0.26226,	0.26326,
	0.26426,	0.26527,	0.26627,	0.26727,
0.26827,	0.26927,	0.27027,	0.27127,	0.27227,
0.27327,	0.27427,	0.27528,	0.27628,	0.27728,
0.27828,	0.27928,	0.28028,	0.28128,	0.28228,
0.28328,	0.28428,	0.28529,	0.28629,	0.28729,

	0.28829,	0.28929,	0.29029,	0.29129,
0.29229,	0.29329,	0.29429,	0.2953,	0.2963,
0.2973,	0.2983,	0.2993,	0.3003,	0.3013,
0.3023,	0.3033,	0.3043,	0.30531,	0.30631,
0.30731,	0.30831,	0.30931,	0.31031,	0.31131,
	0.31231,	0.31331,	0.31431,	0.31532,
0.31632,	0.31732,	0.31832,	0.31932,	0.32032,
0.32132,	0.32232,	0.32332,	0.32432,	0.32533,
0.32633,	0.32733,	0.32833,	0.32933,	0.33033,
0.33133,	0.33233,	0.33333,	0.33433,	0.33534,
	0.33634,	0.33734,	0.33834,	0.33934,
0.34034,	0.34134,	0.34234,	0.34334,	0.34434,
0.34535,	0.34635,	0.34735,	0.34835,	0.34935,
0.35035,	0.35135,	0.35235,	0.35335,	0.35435,
0.35536,	0.35636,	0.35736,	0.35836,	0.35936,
	0.36036,	0.36136,	0.36236,	0.36336,
0.36436,	0.36537,	0.36637,	0.36737,	0.36837,
0.36937,	0.37037,	0.37137,	0.37237,	0.37337,
0.37437,	0.37538,	0.37638,	0.37738,	0.37838,
0.37938,	0.38038,	0.38138,	0.38238,	0.38338,
	0.38438,	0.38539,	0.38639,	0.38739,
0.38839,	0.38939,	0.39039,	0.39139,	0.39239,
0.39339,	0.39439,	0.3954,	0.3964,	0.3974,
0.3984,	0.3994,	0.4004,	0.4014,	0.4024,
0.4034,	0.4044,	0.40541,	0.40641,	0.40741,
	0.40841,	0.40941,	0.41041,	0.41141,
0.41241,	0.41341,	0.41441,	0.41542,	0.41642,
0.41742,	0.41842,	0.41942,	0.42042,	0.42142,
0.42242,	0.42342,	0.42442,	0.42543,	0.42643,
0.42743,	0.42843,	0.42943,	0.43043,	0.43143,
	0.43243,	0.43343,	0.43443,	0.43544,
0.43644,	0.43744,	0.43844,	0.43944,	0.44044,
0.44144,	0.44244,	0.44344,	0.44444,	0.44545,
0.44645,	0.44745,	0.44845,	0.44945,	0.45045,
0.45145,	0.45245,	0.45345,	0.45445,	0.45546,
	0.45646,	0.45746,	0.45846,	0.45946,
0.46046,	0.46146,	0.46246,	0.46346,	0.46446,
0.46547,	0.46647,	0.46747,	0.46847,	0.46947,
0.47047,	0.47147,	0.47247,	0.47347,	0.47447,
0.47548,	0.47648,	0.47748,	0.47848,	0.47948,
	0.48048,	0.48148,	0.48248,	0.48348,
0.48448,	0.48549,	0.48649,	0.48749,	0.48849,
0.48949,	0.49049,	0.49149,	0.49249,	0.49349,
0.49449,	0.4955,	0.4965,	0.4975,	0.4985,
0.4995,	0.5005,	0.5015,	0.5025,	0.5035,
	0.5045,	0.50551,	0.50651,	0.50751,
0.50851,	0.50951,	0.51051,	0.51151,	0.51251,
0.51351,	0.51451,	0.51552,	0.51652,	0.51752,
0.51852,	0.51952,	0.52052,	0.52152,	0.52252,

0.52352,	0.52452,	0.52553,	0.52653,	0.52753,
	0.52853,	0.52953,	0.53053,	0.53153,
0.53253,	0.53353,	0.53453,	0.53554,	0.53654,
0.53754,	0.53854,	0.53954,	0.54054,	0.54154,
0.54254,	0.54354,	0.54454,	0.54555,	0.54655,
0.54755,	0.54855,	0.54955,	0.55055,	0.55155,
	0.55255,	0.55355,	0.55455,	0.55556,
0.55656,	0.55756,	0.55856,	0.55956,	0.56056,
0.56156,	0.56256,	0.56356,	0.56456,	0.56557,
0.56657,	0.56757,	0.56857,	0.56957,	0.57057,
0.57157,	0.57257,	0.57357,	0.57457,	0.57558,
	0.57658,	0.57758,	0.57858,	0.57958,
0.58058,	0.58158,	0.58258,	0.58358,	0.58458,
0.58559,	0.58659,	0.58759,	0.58859,	0.58959,
0.59059,	0.59159,	0.59259,	0.59359,	0.59459,
0.5956,	0.5966,	0.5976,	0.5986,	0.5996,
	0.6006,	0.6016,	0.6026,	0.6036,
0.6046,	0.60561,	0.60661,	0.60761,	0.60861,
0.60961,	0.61061,	0.61161,	0.61261,	0.61361,
0.61461,	0.61562,	0.61662,	0.61762,	0.61862,
0.61962,	0.62062,	0.62162,	0.62262,	0.62362,
	0.62462,	0.62563,	0.62663,	0.62763,
0.62863,	0.62963,	0.63063,	0.63163,	0.63263,
0.63363,	0.63463,	0.63564,	0.63664,	0.63764,
0.63864,	0.63964,	0.64064,	0.64164,	0.64264,
0.64364,	0.64464,	0.64565,	0.64665,	0.64765,
	0.64865,	0.64965,	0.65065,	0.65165,
0.65265,	0.65365,	0.65465,	0.65566,	0.65666,
0.65766,	0.65866,	0.65966,	0.66066,	0.66166,
0.66266,	0.66366,	0.66466,	0.66567,	0.66667,
0.66767,	0.66867,	0.66967,	0.67067,	0.67167,
	0.67267,	0.67367,	0.67467,	0.67568,
0.67668,	0.67768,	0.67868,	0.67968,	0.68068,
0.68168,	0.68268,	0.68368,	0.68468,	0.68569,
0.68669,	0.68769,	0.68869,	0.68969,	0.69069,
0.69169,	0.69269,	0.69369,	0.69469,	0.6957,
	0.6967,	0.6977,	0.6987,	0.6997,
0.7007,	0.7017,	0.7027,	0.7037,	0.7047,
0.70571,	0.70671,	0.70771,	0.70871,	0.70971,
0.71071,	0.71171,	0.71271,	0.71371,	0.71471,
0.71572,	0.71672,	0.71772,	0.71872,	0.71972,
	0.72072,	0.72172,	0.72272,	0.72372,
0.72472,	0.72573,	0.72673,	0.72773,	0.72873,
0.72973,	0.73073,	0.73173,	0.73273,	0.73373,
0.73473,	0.73574,	0.73674,	0.73774,	0.73874,
0.73974,	0.74074,	0.74174,	0.74274,	0.74374,
	0.74474,	0.74575,	0.74675,	0.74775,
0.74875,	0.74975,	0.75075,	0.75175,	0.75275,
0.75375,	0.75475,	0.75576,	0.75676,	0.75776,

0.75876,	0.75976,	0.76076,	0.76176,	0.76276,
0.76376,	0.76476,	0.76577,	0.76677,	0.76777,
	0.76877,	0.76977,	0.77077,	0.77177,
0.77277,	0.77377,	0.77477,	0.77578,	0.77678,
0.77778,	0.77878,	0.77978,	0.78078,	0.78178,
0.78278,	0.78378,	0.78478,	0.78579,	0.78679,
0.78779,	0.78879,	0.78979,	0.79079,	0.79179,
	0.79279,	0.79379,	0.79479,	0.7958,
0.7968,	0.7978,	0.7988,	0.7998,	0.8008,
0.8018,	0.8028,	0.8038,	0.8048,	0.80581,
0.80681,	0.80781,	0.80881,	0.80981,	0.81081,
0.81181,	0.81281,	0.81381,	0.81481,	0.81582,
	0.81682,	0.81782,	0.81882,	0.81982,
0.82082,	0.82182,	0.82282,	0.82382,	0.82482,
0.82583,	0.82683,	0.82783,	0.82883,	0.82983,
0.83083,	0.83183,	0.83283,	0.83383,	0.83483,
0.83584,	0.83684,	0.83784,	0.83884,	0.83984,
	0.84084,	0.84184,	0.84284,	0.84384,
0.84484,	0.84585,	0.84685,	0.84785,	0.84885,
0.84985,	0.85085,	0.85185,	0.85285,	0.85385,
0.85485,	0.85586,	0.85686,	0.85786,	0.85886,
0.85986,	0.86086,	0.86186,	0.86286,	0.86386,
	0.86486,	0.86587,	0.86687,	0.86787,
0.86887,	0.86987,	0.87087,	0.87187,	0.87287,
0.87387,	0.87487,	0.87588,	0.87688,	0.87788,
0.87888,	0.87988,	0.88088,	0.88188,	0.88288,
0.88388,	0.88488,	0.88589,	0.88689,	0.88789,
	0.88889,	0.88989,	0.89089,	0.89189,
0.89289,	0.89389,	0.89489,	0.8959,	0.8969,
0.8979,	0.8989,	0.8999,	0.9009,	0.9019,
0.9029,	0.9039,	0.9049,	0.90591,	0.90691,
0.90791,	0.90891,	0.90991,	0.91091,	0.91191,
	0.91291,	0.91391,	0.91491,	0.91592,
0.91692,	0.91792,	0.91892,	0.91992,	0.92092,
0.92192,	0.92292,	0.92392,	0.92492,	0.92593,
0.92693,	0.92793,	0.92893,	0.92993,	0.93093,
0.93193,	0.93293,	0.93393,	0.93493,	0.93594,
	0.93694,	0.93794,	0.93894,	0.93994,
0.94094,	0.94194,	0.94294,	0.94394,	0.94494,
0.94595,	0.94695,	0.94795,	0.94895,	0.94995,
0.95095,	0.95195,	0.95295,	0.95395,	0.95495,
0.95596,	0.95696,	0.95796,	0.95896,	0.95996,
	0.96096,	0.96196,	0.96296,	0.96396,
0.96496,	0.96597,	0.96697,	0.96797,	0.96897,
0.96997,	0.97097,	0.97197,	0.97297,	0.97397,
0.97497,	0.97598,	0.97698,	0.97798,	0.97898,
0.97998,	0.98098,	0.98198,	0.98298,	0.98398,
	0.98498,	0.98599,	0.98699,	0.98799,
0.98899,	0.98999,	0.99099,	0.99199,	0.99299,

0.99399,	0.99499,	0.996,	0.997,	0.998,	
0.999,	1]],	array([[	0.93475,	0.93475,	0.92418,
0.91673,	0.91015,	0.90376,	0.8981,	0.89365,	
0.88935,	0.88521,	0.88106,	0.87821,	0.87521,	
0.8724,	0.86966,	0.86715,	0.86437,	0.86186,	
0.85981,	0.85722,	0.85513,	0.8527,	0.84996,	
	0.84749,	0.84551,	0.84418,	0.84209,	
0.84038,	0.83875,	0.83703,	0.83548,	0.83392,	
0.83232,	0.83026,	0.8281,	0.8262,	0.82449,	
0.82293,	0.82156,	0.81977,	0.81848,	0.81646,	
0.81468,	0.81281,	0.81148,	0.81019,		
	0.80882,	0.80802,	0.80627,	0.80506,	
0.80369,	0.80194,	0.80046,	0.79981,	0.79867,	
0.79779,	0.79612,	0.79532,	0.79364,	0.79251,	
0.7916,	0.78992,	0.78871,	0.78757,	0.78643,	
0.78504,	0.78403,	0.78289,	0.78118,		
	0.7803,	0.77924,	0.7781,	0.77711,	
0.77589,	0.77445,	0.77297,	0.77224,	0.77084,	
0.76947,	0.76862,	0.76715,	0.76608,	0.76521,	
0.76392,	0.76285,	0.76144,	0.76055,	0.75947,	
0.75795,	0.75681,	0.75532,	0.75441,		
	0.75331,	0.75221,	0.7511,	0.75023,	
0.74932,	0.74806,	0.74681,	0.74627,	0.74471,	
0.74361,	0.74274,	0.74156,	0.74095,	0.73966,	
0.73852,	0.7376,	0.73665,	0.73567,	0.73441,	
0.73342,	0.73255,	0.7319,	0.73089,		
	0.72989,	0.72878,	0.72802,	0.72692,	
0.72643,	0.72532,	0.72452,	0.72384,	0.72252,	
0.72175,	0.72011,	0.71916,	0.71806,	0.71703,	
0.71605,	0.71551,	0.71441,	0.71304,	0.71209,	
0.71125,	0.71051,	0.70924,	0.70837,		
	0.70745,	0.70669,	0.70551,	0.70468,	
0.70373,	0.70312,	0.70228,	0.70084,	0.70015,	
0.69939,	0.69878,	0.69795,	0.69711,	0.69624,	
0.69536,	0.69483,	0.69388,	0.69285,	0.6924,	
0.69167,	0.69118,	0.69019,	0.68932,		
	0.68867,	0.68817,	0.68698,	0.68589,	
0.68525,	0.68414,	0.68373,	0.68274,	0.68156,	
0.68065,	0.67985,	0.67901,	0.67859,	0.67787,	
0.67688,	0.67608,	0.6757,	0.67456,	0.67354,	
0.6724,	0.67171,	0.67072,	0.66992,		
	0.66985,	0.66875,	0.66773,	0.66673,	
0.66635,	0.66552,	0.66449,	0.66385,	0.66281,	
0.66212,	0.66183,	0.66118,	0.66,	0.65967,	
0.65837,	0.6575,	0.65665,	0.65544,	0.65494,	
0.65418,	0.65376,	0.65228,	0.65202,		
	0.65124,	0.65053,	0.65007,	0.64886,	
0.64856,	0.6473,	0.64665,	0.6454,	0.64494,	
0.64395,	0.64346,	0.64202,	0.64156,	0.64046,	

0.63992,	0.63875,	0.63818,	0.63738,	0.63681,
0.63548,	0.63513,	0.63399,	0.63323,	
	0.6324,	0.63163,	0.63106,	0.62996,
0.62939,	0.62828,	0.62757,	0.62646,	0.62595,
0.62517,	0.62354,	0.6229,	0.62186,	0.62091,
0.6202,	0.61909,	0.61856,	0.61776,	0.61664,
0.61635,	0.61532,	0.61456,	0.61392,	
	0.61281,	0.61228,	0.61171,	0.61071,
0.61004,	0.60958,	0.60794,	0.60726,	0.60635,
0.60578,	0.6051,	0.60441,	0.603,	0.60248,
0.60175,	0.60068,	0.6,	0.59928,	0.59802,
0.597,	0.5965,	0.59586,	0.5943,	
	0.59335,	0.59275,	0.59202,	0.59061,
0.5899,	0.58943,	0.58802,	0.587,	0.58654,
0.58608,	0.58456,	0.58399,	0.58365,	0.58297,
0.58179,	0.58087,	0.58024,	0.57962,	0.57814,
0.57768,	0.57682,	0.57601,	0.57506,	
	0.57395,	0.57357,	0.57293,	0.57224,
0.5706,	0.56989,	0.56901,	0.56842,	0.5678,
0.56658,	0.56603,	0.56544,	0.56471,	0.56319,
0.56274,	0.56196,	0.56118,	0.56015,	0.55916,
0.55856,	0.55718,	0.55631,	0.55578,	
	0.55529,	0.55464,	0.55338,	0.55296,
0.55221,	0.55167,	0.55053,	0.54954,	0.54825,
0.5476,	0.54707,	0.54653,	0.54618,	0.54525,
0.54471,	0.54316,	0.54278,	0.5419,	0.54118,
0.5403,	0.53978,	0.53844,	0.53772,	
	0.53713,	0.53639,	0.53576,	0.53508,
0.53452,	0.53316,	0.53236,	0.53152,	0.5308,
0.53011,	0.52958,	0.52918,	0.52859,	0.52779,
0.52703,	0.52532,	0.52429,	0.52333,	0.5223,
0.52118,	0.52057,	0.52017,	0.5192,	
	0.51867,	0.51787,	0.51635,	0.51562,
0.5144,	0.51365,	0.51264,	0.51184,	0.51131,
0.51011,	0.5089,	0.50787,	0.50665,	0.50573,
0.50474,	0.50407,	0.5031,	0.50259,	0.50195,
0.50092,	0.4999,	0.49916,	0.49749,	
	0.49635,	0.49528,	0.49445,	0.49361,
0.49303,	0.49217,	0.49097,	0.49019,	0.48909,
0.48826,	0.4875,	0.487,	0.48609,	0.4851,
0.48422,	0.4835,	0.48151,	0.48076,	0.47999,
0.47931,	0.47844,	0.47802,	0.47691,	
	0.47614,	0.47473,	0.47388,	0.47285,
0.47172,	0.47088,	0.4699,	0.46916,	0.46859,
0.4668,	0.466,	0.46505,	0.46418,	0.46322,
0.46205,	0.46113,	0.46026,	0.45938,	0.45854,
0.45759,	0.45649,	0.45551,	0.45442,	
	0.45357,	0.45266,	0.45202,	0.45119,
0.45038,	0.4497,	0.44878,	0.44798,	0.44723,
0.44627,	0.44525,	0.44449,	0.44357,	0.44186,

0.44076,	0.44004,	0.43909,	0.43848,	0.43745,
0.43641,	0.43527,	0.43464,	0.4339,	
	0.43308,	0.43234,	0.43123,	0.43038,
0.42935,	0.42856,	0.42779,	0.42659,	0.42589,
0.42541,	0.42446,	0.42369,	0.42316,	0.42232,
0.42149,	0.42068,	0.41973,	0.41882,	0.41795,
0.4173,	0.41616,	0.41525,	0.41418,	
	0.41335,	0.4124,	0.41144,	0.41064,
0.40961,	0.40893,	0.40825,	0.40729,	0.4062,
0.40524,	0.40421,	0.40333,	0.4024,	0.40143,
0.40036,	0.39949,	0.39856,	0.3977,	0.39677,
0.39597,	0.39494,	0.39386,	0.3926,	
	0.3921,	0.39138,	0.3903,	0.38962,
0.38875,	0.38787,	0.38716,	0.38617,	0.38514,
0.38392,	0.38312,	0.38209,	0.38118,	0.38019,
0.37863,	0.37791,	0.37673,	0.37586,	0.37471,
0.37395,	0.37315,	0.37201,	0.3711,	
	0.36999,	0.36915,	0.36847,	0.36782,
0.36679,	0.36577,	0.36497,	0.36436,	0.36314,
0.36223,	0.36131,	0.3604,	0.35918,	0.35785,
0.3569,	0.35606,	0.35492,	0.35378,	0.3529,
0.35205,	0.35123,	0.35001,	0.34906,	
	0.34818,	0.34711,	0.34578,	0.3448,
0.34384,	0.34183,	0.34078,	0.33956,	0.33842,
0.33739,	0.33639,	0.33548,	0.33435,	0.33317,
0.33203,	0.33099,	0.32967,	0.32853,	0.32758,
0.3264,	0.3251,	0.32381,	0.32301,	
	0.32156,	0.32065,	0.31962,	0.31886,
0.31757,	0.31509,	0.3141,	0.31304,	0.3118,
0.31085,	0.31001,	0.30906,	0.3083,	0.30708,
0.30583,	0.30452,	0.3032,	0.30191,	0.30053,
0.29962,	0.29841,	0.29633,	0.29534,	
	0.29416,	0.29319,	0.29234,	0.29112,
0.29008,	0.28907,	0.28802,	0.28693,	0.2856,
0.28307,	0.28224,	0.28113,	0.2801,	0.27896,
0.27759,	0.27641,	0.2754,	0.27421,	0.27343,
0.27228,	0.27114,	0.27019,	0.26806,	
	0.26684,	0.26551,	0.26445,	0.26345,
0.26262,	0.26123,	0.26016,	0.25925,	0.25796,
0.25689,	0.25556,	0.2539,	0.25299,	0.25189,
0.25082,	0.25002,	0.24882,	0.24787,	0.24684,
0.2446,	0.24376,	0.24285,	0.24163,	
	0.24055,	0.23948,	0.23856,	0.2375,
0.23659,	0.23474,	0.23342,	0.23227,	0.23136,
0.2303,	0.22917,	0.22761,	0.2267,	0.2254,
0.22318,	0.2219,	0.22104,	0.21998,	0.21787,
0.21688,	0.21566,	0.21509,	0.21375,	
	0.21313,	0.21144,	0.21119,	0.20904,
0.20756,	0.20703,	0.20541,	0.20487,	0.20365,

0.20293,	0.20076,	0.1993,	0.19857,	0.19654,
0.19528,	0.19456,	0.193,	0.19228,	0.19096,
0.18992,	0.18714,	0.18547,	0.18462,	
	0.18297,	0.18214,	0.18024,	0.17805,
0.17694,	0.17476,	0.17384,	0.17225,	0.16964,
0.16865,	0.1657,	0.16376,	0.163,	0.16157,
0.161,	0.15927,	0.15752,	0.15683,	0.1556,
0.1548,	0.1528,	0.15144,	0.15108,	
	0.15008,	0.14932,	0.14776,	0.14677,
0.14624,	0.14506,	0.14371,	0.14317,	0.14171,
0.14102,	0.14059,	0.13952,	0.13829,	0.1379,
0.13686,	0.13539,	0.13486,	0.13386,	0.13316,
0.1317,	0.13039,	0.12977,	0.12898,	
	0.12755,	0.12689,	0.12555,	0.12371,
0.12205,	0.12152,	0.12047,	0.11954,	0.11752,
0.11641,	0.11606,	0.11402,	0.11291,	0.11214,
0.11029,	0.10899,	0.10841,	0.10675,	0.1059,
0.10537,	0.10354,	0.10244,	0.10116,	
	0.10068,	0.09901,	0.097736,	0.095891,
0.095133,	0.09357,	0.092613,	0.092129,	0.090259,
0.089071,	0.087481,	0.087166,	0.086624,	0.085387,
0.084765,	0.083446,	0.082556,	0.082056,	0.080993,
0.07981,	0.078327,	0.076995,	0.076443,	
	0.074748,	0.073742,	0.071968,	0.071382,
0.069645,	0.068415,	0.066675,	0.06636,	0.064846,
0.063658,	0.062409,	0.062135,	0.060846,	0.059586,
0.057915,	0.056274,	0.055858,	0.054088,	0.052964,
0.051139,	0.050489,	0.048928,	0.048417,	
	0.047023,	0.046115,	0.045575,	0.044297,
0.04403,	0.042983,	0.041966,	0.041154,	0.040164,
0.039416,	0.039133,	0.038371,	0.037588,	0.036922,
0.03597,	0.03479,	0.034141,	0.033151,	0.031818,
0.031079,	0.029837,	0.028276,	0.026571,	
	0.025497,	0.023974,	0.022717,	0.021536,
0.021109,	0.01986,	0.018831,	0.018107,	0.017573,
0.016979,	0.01673,	0.016464,	0.015854,	0.015244,
0.0149,	0.014441,	0.014021,	0.013531,	0.013219,
0.012799,	0.012112,	0.011654,	0.011234,	
	0.010852,	0.01047,	0.010126,	0.0099718,
0.0097037,	0.0092075,	0.0088634,	0.0085573,	0.0080611,
0.0075649,	0.0072588,	0.0069908,	0.0066847,	0.0062265,
0.0061485,	0.0059565,	0.0056504,	0.0051542,	0.0049242,
0.0047615,	0.0045782,	0.0041961,	0.0038139,	
	0.0035078,	0.0032778,	0.0030478,	0.0029298,
0.0028158,	0.0024337,	0.0022797,	0.0021257,	0.0020315,
0.0017417,	0.0016113,	0.0015097,	0.0013623,	0.0012778,
0.0011893,	0.0011503,	0.0010163,	0.00092796,	0.00084326,
0.00070923,	0.00065609,	0.00061227,	0.00044021,	
	0.00038422,	0.00034325,	0.0002472,	0.00022642,


```
'val_batch0_pred.jpg',
'R_curve.png',
'events.out.tfevents.1741455867.6cd5553c5587.31.0',
'val_batch2_pred.jpg',
'args.yaml',
'train_batch2.jpg',
'P_curve.png',
'PR_curve.png',
'train_batch0.jpg',
'confusion_matrix.png',
'F1_curve.png',
'results.csv',
'val_batch0_labels.jpg']
```

```
results_csv = os.path.join(results_path, "results.csv")
df_results = pd.read_csv(results_csv)
print(df_results.head())
```

	epoch	time	train/box_loss	train/cls_loss	train/df_l_loss	\
0	1	424.203	1.73733	2.27427	1.28781	
1	2	835.029	1.70722	1.78509	1.29986	
2	3	1240.160	1.65339	1.64805	1.26459	
3	4	1642.310	1.58422	1.51174	1.23188	
4	5	2044.560	1.49633	1.36960	1.18846	

	metrics/precision(B)	metrics/recall(B)	metrics/mAP50(B)	\
0	0.34233	0.40042	0.30674	
1	0.51577	0.46113	0.44771	
2	0.55060	0.52860	0.51236	
3	0.60303	0.57332	0.58552	
4	0.65815	0.60650	0.64120	

	metrics/mAP50-95(B)	val/box_loss	val/cls_loss	val/df_l_loss
lr/pg0 \				
0	0.16786	1.87488	2.09891	1.33479
0.000666				
1	0.24672	1.77922	1.76855	1.28747
0.001069				
2	0.29161	1.69503	1.58848	1.25266
0.001208				
3	0.34027	1.64751	1.44037	1.22026
0.000812				
4	0.37994	1.55979	1.31872	1.17336
0.000416				

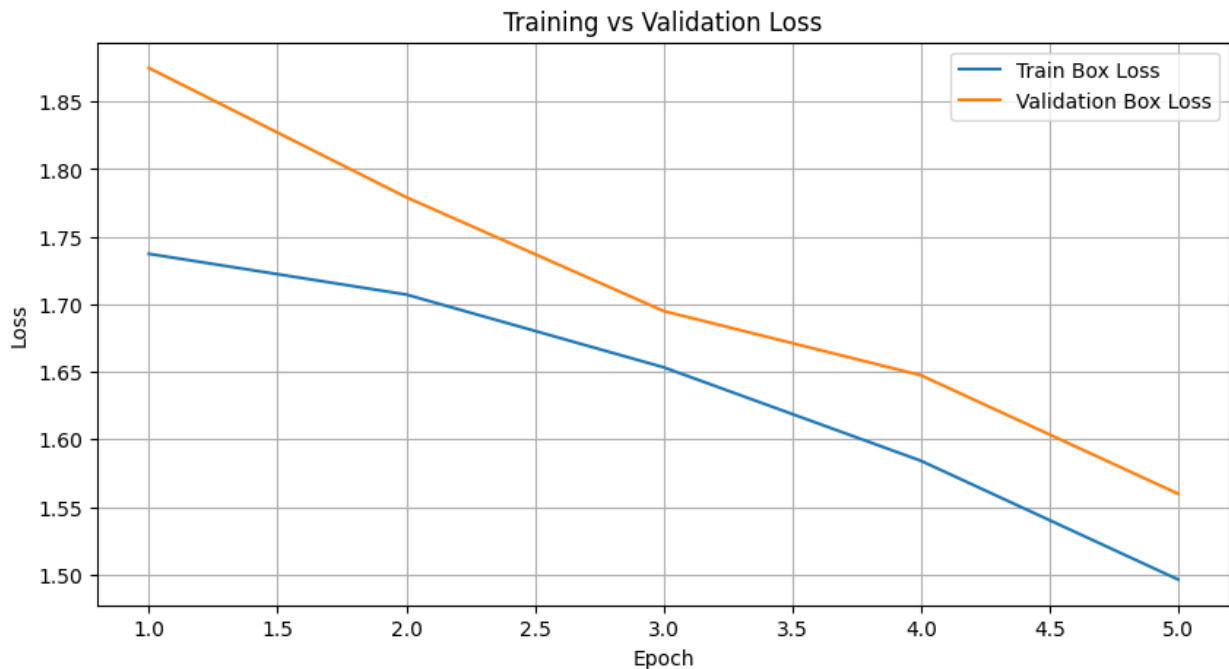
	lr/pg1	lr/pg2
0	0.000666	0.000666
1	0.001069	0.001069
2	0.001208	0.001208


```
3  0.000812  0.000812
4  0.000416  0.000416
```

```
print("Available Columns:", df_results.columns)
```

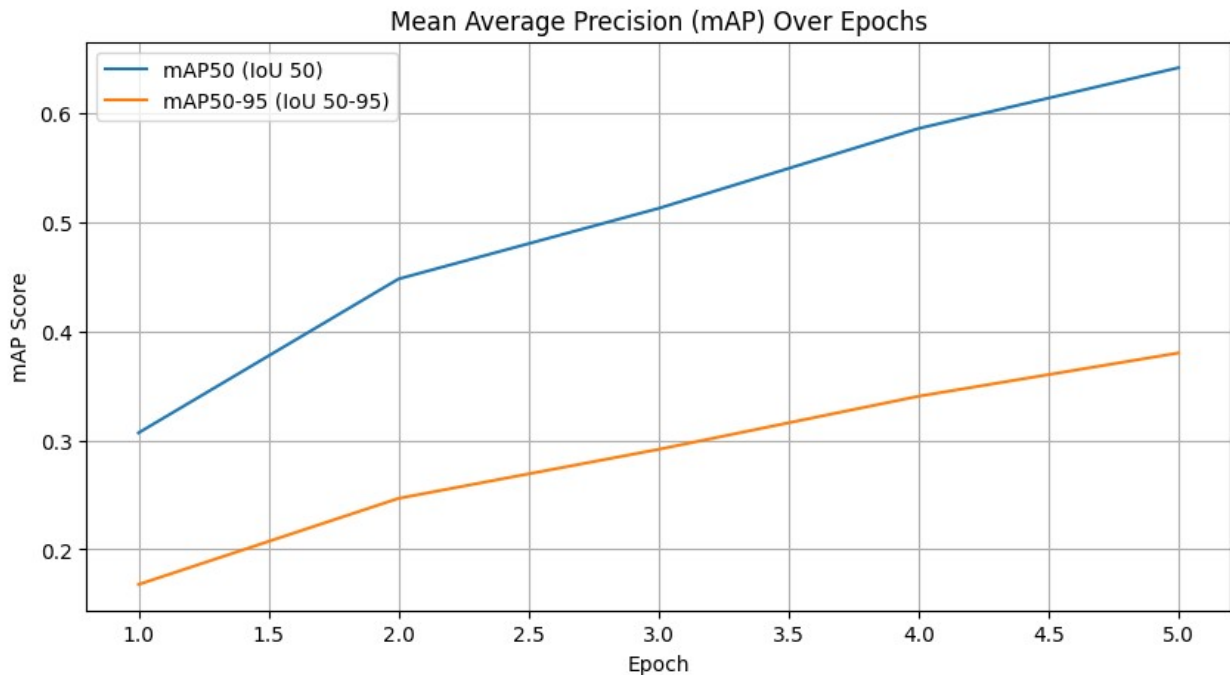
```
Available Columns: Index(['epoch', 'time', 'train/box_loss',
                           'train/cls_loss', 'train/dfl_loss',
                           'metrics/precision(B)', 'metrics/recall(B)',
                           'metrics/mAP50(B)',
                           'metrics/mAP50-95(B)', 'val/box_loss', 'val/cls_loss',
                           'val/dfl_loss',
                           'lr/pg0', 'lr/pg1', 'lr/pg2'],
                          dtype='object')
```

```
plt.figure(figsize=(10, 5))
plt.plot(df_results["epoch"], df_results["train/box_loss"],
         label="Train Box Loss")
plt.plot(df_results["epoch"], df_results["val/box_loss"],
         label="Validation Box Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Training vs Validation Loss")
plt.legend()
plt.grid()
plt.show()
```

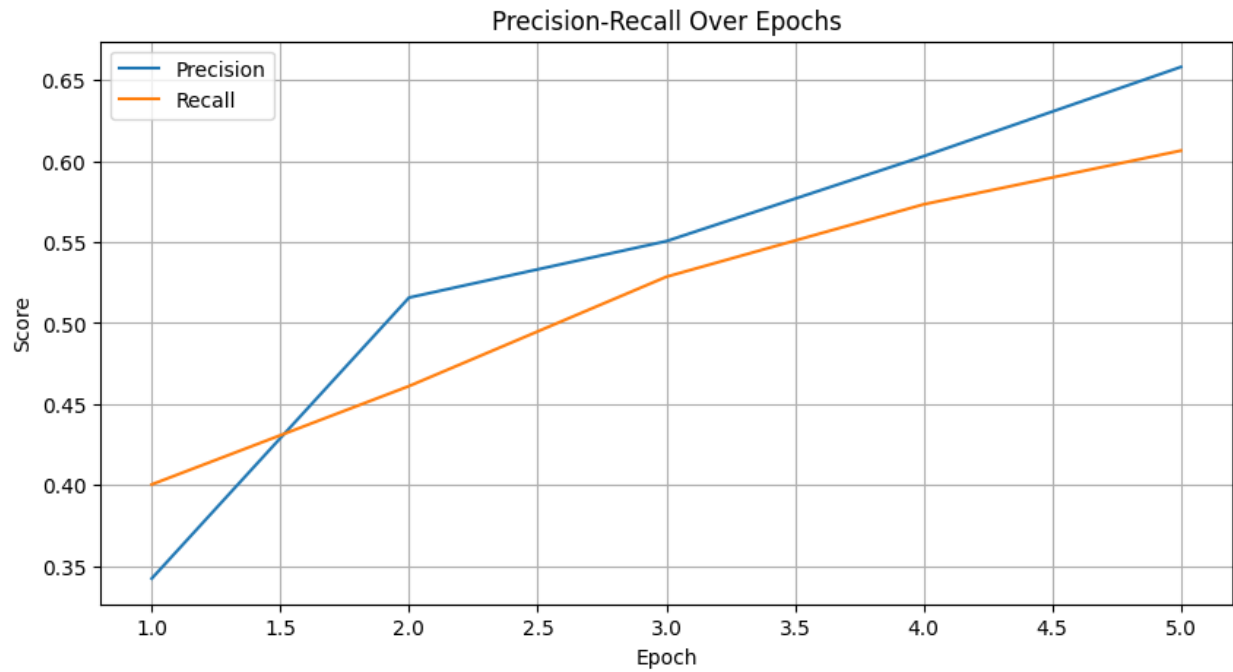


```
plt.figure(figsize=(10, 5))
plt.plot(df_results["epoch"], df_results["metrics/mAP50(B)"],
         label="mAP50 (IoU 50)")
```

```
plt.plot(df_results["epoch"], df_results["metrics/mAP50-95(B)"],
label="mAP50-95 (IoU 50-95)")
plt.xlabel("Epoch")
plt.ylabel("mAP Score")
plt.title("Mean Average Precision (mAP) Over Epochs")
plt.legend()
plt.grid()
plt.show()
```

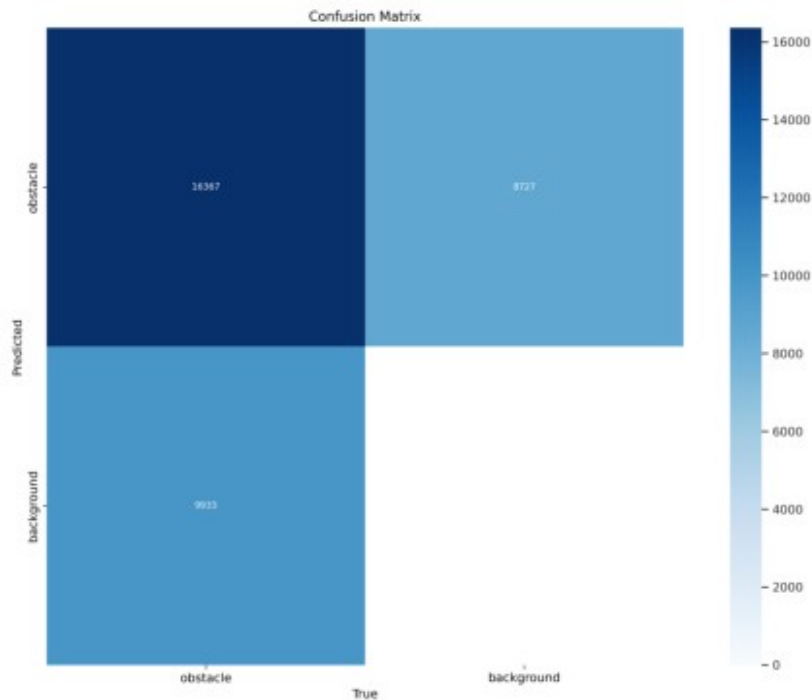


```
plt.figure(figsize=(10, 5))
plt.plot(df_results["epoch"], df_results["metrics/precision(B)"],
label="Precision")
plt.plot(df_results["epoch"], df_results["metrics/recall(B)"],
label="Recall")
plt.xlabel("Epoch")
plt.ylabel("Score")
plt.title("Precision-Recall Over Epochs")
plt.legend()
plt.grid()
plt.show()
```



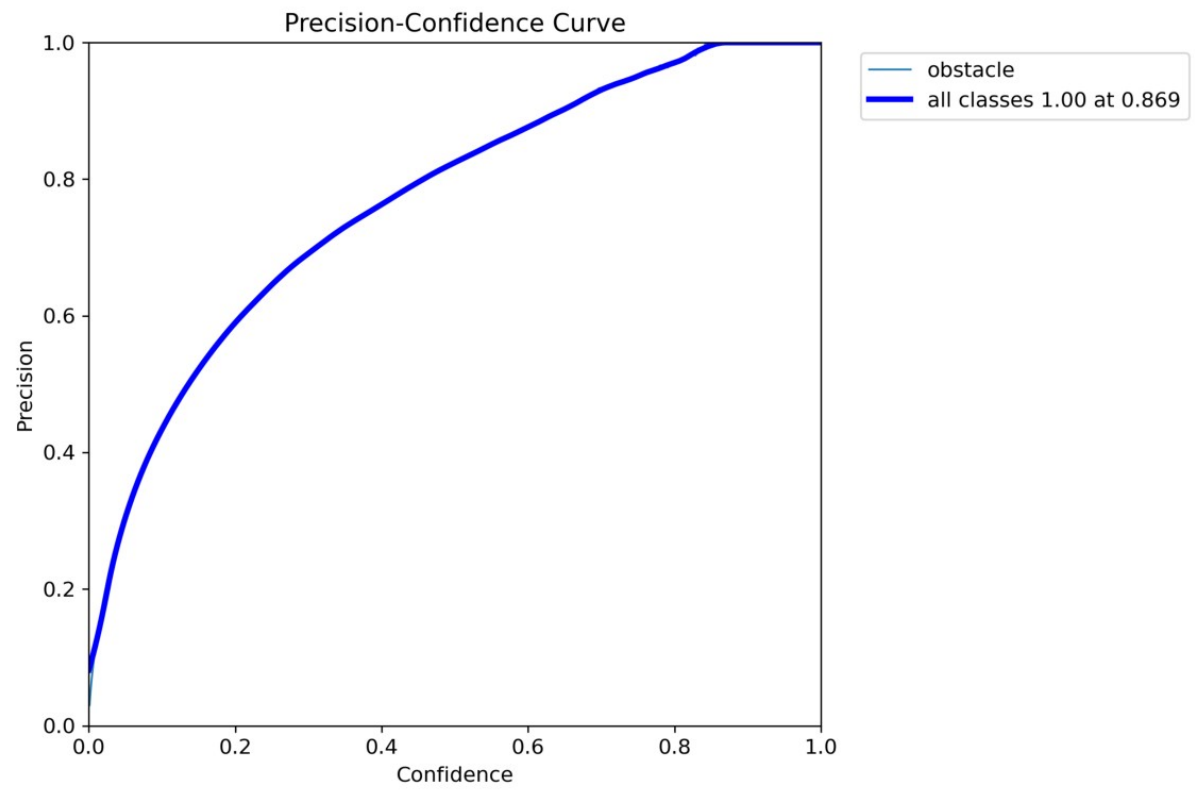
```
conf_matrix_path =  
"/kaggle/working/runs/detect/yolo_v8_model/confusion_matrix.png"  
conf_matrix = cv2.imread(conf_matrix_path)  
plt.imshow(cv2.cvtColor(conf_matrix, cv2.COLOR_BGR2RGB))  
plt.title("Confusion Matrix")  
plt.axis("off")  
plt.show()
```

Confusion Matrix

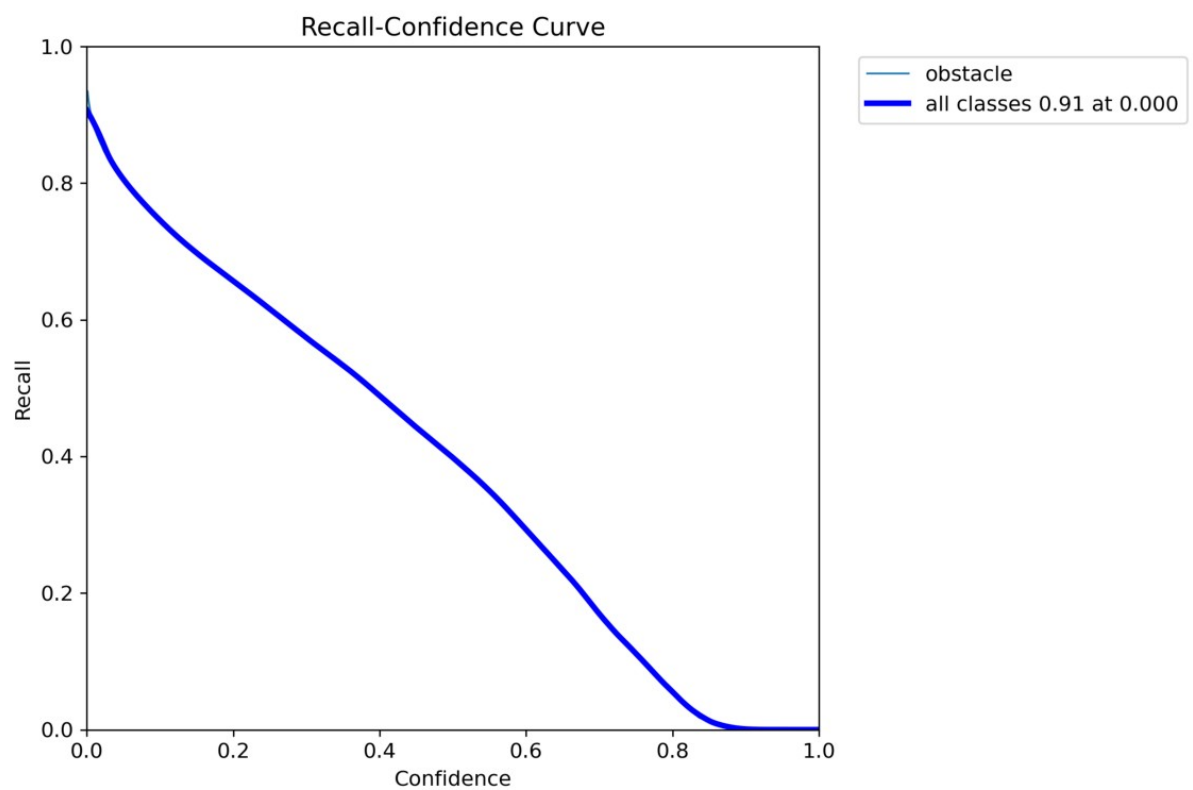


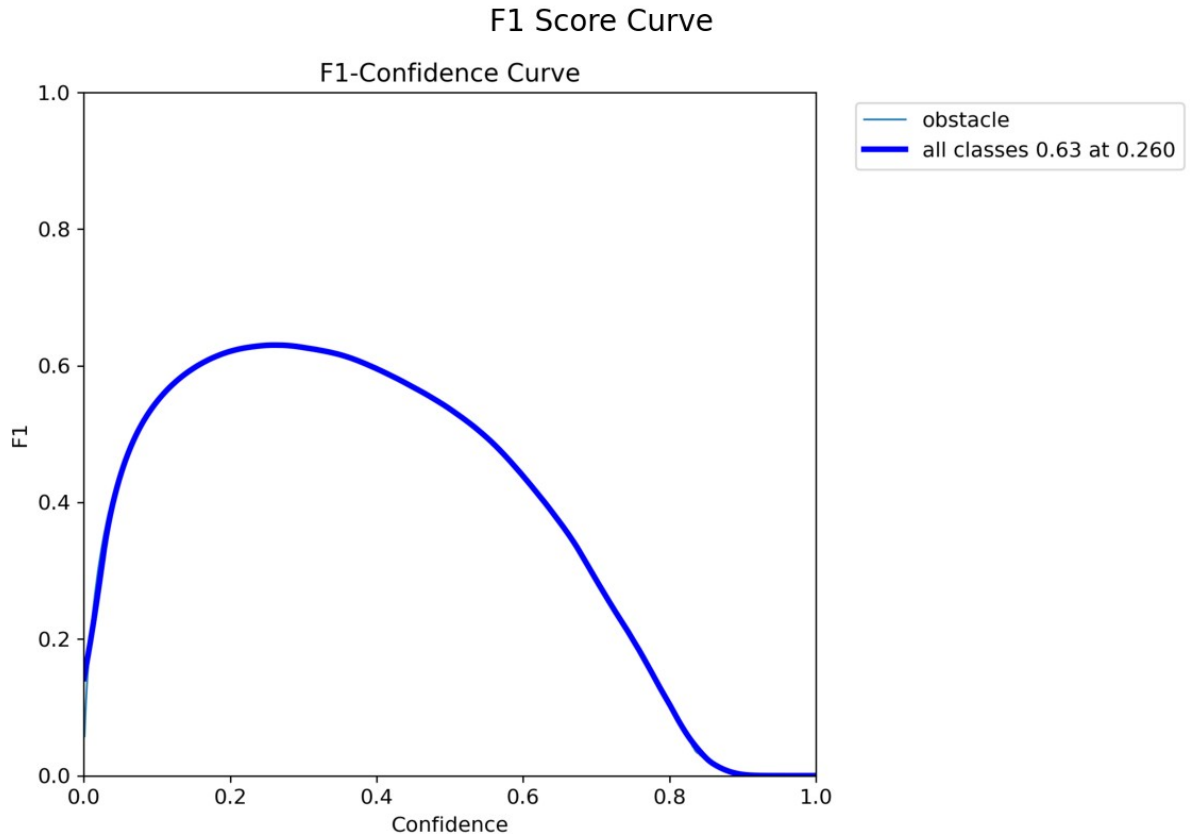
```
precision_curve =  
cv2.imread("/kaggle/working/runs/detect/yolo_v8_model/P_curve.png")  
recall_curve =  
cv2.imread("/kaggle/working/runs/detect/yolo_v8_model/R_curve.png")  
f1_curve =  
cv2.imread("/kaggle/working/runs/detect/yolo_v8_model/F1_curve.png")  
  
def show_large_image(image, title):  
    plt.figure(figsize=(15, 10)) # Bigger image  
    plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))  
    plt.title(title, fontsize=20)  
    plt.axis("off")  
    plt.show()  
show_large_image(precision_curve, "Precision Curve")  
show_large_image(recall_curve, "Recall Curve")  
show_large_image(f1_curve, "F1 Score Curve")
```

Precision Curve



Recall Curve





#####FUSING#####

```
!pip install ultralytics opencv-python numpy ffmpeg-python
```

```
Requirement already satisfied: ultralytics in  
/usr/local/lib/python3.10/dist-packages (8.3.85)  
Requirement already satisfied: opencv-python in  
/usr/local/lib/python3.10/dist-packages (4.10.0.84)  
Requirement already satisfied: numpy in  
/usr/local/lib/python3.10/dist-packages (1.26.4)  
Requirement already satisfied: ffmpeg-python in  
/usr/local/lib/python3.10/dist-packages (0.2.0)  
Requirement already satisfied: matplotlib>=3.3.0 in  
/usr/local/lib/python3.10/dist-packages (from ultralytics) (3.7.5)  
Requirement already satisfied: pillow>=7.1.2 in  
/usr/local/lib/python3.10/dist-packages (from ultralytics) (11.0.0)  
Requirement already satisfied: pyyaml>=5.3.1 in  
/usr/local/lib/python3.10/dist-packages (from ultralytics) (6.0.2)  
Requirement already satisfied: requests>=2.23.0 in  
/usr/local/lib/python3.10/dist-packages (from ultralytics) (2.32.3)  
Requirement already satisfied: scipy>=1.4.1 in  
/usr/local/lib/python3.10/dist-packages (from ultralytics) (1.13.1)  
Requirement already satisfied: torch>=1.8.0 in  
/usr/local/lib/python3.10/dist-packages (from ultralytics)
```

(2.5.1+cu121)
Requirement already satisfied: torchvision>=0.9.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics)
(0.20.1+cu121)
Requirement already satisfied: tqdm>=4.64.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (4.67.1)
Requirement already satisfied: psutil in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (5.9.5)
Requirement already satisfied: py-cpuinfo in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (9.0.0)
Requirement already satisfied: pandas>=1.1.4 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (2.2.3)
Requirement already satisfied: seaborn>=0.11.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (0.12.2)
Requirement already satisfied: ultralytics-thop>=2.0.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (2.0.14)
Requirement already satisfied: mkl_fft in
/usr/local/lib/python3.10/dist-packages (from numpy) (1.3.8)
Requirement already satisfied: mkl_random in
/usr/local/lib/python3.10/dist-packages (from numpy) (1.2.4)
Requirement already satisfied: mkl_umath in
/usr/local/lib/python3.10/dist-packages (from numpy) (0.1.1)
Requirement already satisfied: mkl in /usr/local/lib/python3.10/dist-
packages (from numpy) (2025.0.1)
Requirement already satisfied: tbb4py in
/usr/local/lib/python3.10/dist-packages (from numpy) (2022.0.0)
Requirement already satisfied: mkl-service in
/usr/local/lib/python3.10/dist-packages (from numpy) (2.4.1)
Requirement already satisfied: future in
/usr/local/lib/python3.10/dist-packages (from ffmpeg-python) (1.0.0)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (1.3.1)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (4.55.3)
Requirement already satisfied: kiwisolver>=1.0.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (1.4.7)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (24.2)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-
>ultralytics) (3.2.0)
Requirement already satisfied: python-dateutil>=2.7 in


```
/usr/local/lib/python3.10/dist-packages (from matplotlib>=3.3.0-  
>ultralitics) (2.9.0.post0)  
Requirement already satisfied: pytz>=2020.1 in  
/usr/local/lib/python3.10/dist-packages (from pandas>=1.1.4-  
>ultralitics) (2025.1)  
Requirement already satisfied: tzdata>=2022.7 in  
/usr/local/lib/python3.10/dist-packages (from pandas>=1.1.4-  
>ultralitics) (2025.1)  
Requirement already satisfied: charset-normalizer<4,>=2 in  
/usr/local/lib/python3.10/dist-packages (from requests>=2.23.0-  
>ultralitics) (3.4.1)  
Requirement already satisfied: idna<4,>=2.5 in  
/usr/local/lib/python3.10/dist-packages (from requests>=2.23.0-  
>ultralitics) (3.10)  
Requirement already satisfied: urllib3<3,>=1.21.1 in  
/usr/local/lib/python3.10/dist-packages (from requests>=2.23.0-  
>ultralitics) (2.3.0)  
Requirement already satisfied: certifi>=2017.4.17 in  
/usr/local/lib/python3.10/dist-packages (from requests>=2.23.0-  
>ultralitics) (2025.1.31)  
Requirement already satisfied: filelock in  
/usr/local/lib/python3.10/dist-packages (from torch>=1.8.0-  
>ultralitics) (3.17.0)  
Requirement already satisfied: typing-extensions>=4.8.0 in  
/usr/local/lib/python3.10/dist-packages (from torch>=1.8.0-  
>ultralitics) (4.12.2)  
Requirement already satisfied: networkx in  
/usr/local/lib/python3.10/dist-packages (from torch>=1.8.0-  
>ultralitics) (3.4.2)  
Requirement already satisfied: jinja2 in  
/usr/local/lib/python3.10/dist-packages (from torch>=1.8.0-  
>ultralitics) (3.1.4)  
Requirement already satisfied: fsspec in  
/usr/local/lib/python3.10/dist-packages (from torch>=1.8.0-  
>ultralitics) (2024.12.0)  
Requirement already satisfied: sympy==1.13.1 in  
/usr/local/lib/python3.10/dist-packages (from torch>=1.8.0-  
>ultralitics) (1.13.1)  
Requirement already satisfied: mpmath<1.4,>=1.1.0 in  
/usr/local/lib/python3.10/dist-packages (from sympy==1.13.1-  
>torch>=1.8.0->ultralitics) (1.3.0)  
Requirement already satisfied: intel-openmp>=2024 in  
/usr/local/lib/python3.10/dist-packages (from mkl->numpy) (2024.2.0)  
Requirement already satisfied: tbb==2022.* in  
/usr/local/lib/python3.10/dist-packages (from mkl->numpy) (2022.0.0)  
Requirement already satisfied: tcmlib==1.* in  
/usr/local/lib/python3.10/dist-packages (from tbb==2022.*->mkl->numpy)  
(1.2.0)  
Requirement already satisfied: intel-cmplr-lib-rt in
```

```
/usr/local/lib/python3.10/dist-packages (from mkl_umath->numpy)
(2024.2.0)
Requirement already satisfied: intel-cmplr-lib-ur==2024.2.0 in
/usr/local/lib/python3.10/dist-packages (from intel-openmp>=2024->mkl-
>numpy) (2024.2.0)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.10/dist-packages (from python-dateutil>=2.7-
>matplotlib>=3.3.0->ultralitics) (1.17.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.10/dist-packages (from jinja2->torch>=1.8.0-
>ultralitics) (3.0.2)
```

```
model =
YOLO("/kaggle/working/runs/detect/yolo_v8_model/weights/best.pt")
```

```
input_video_path = "/kaggle/input/test-vid/0000f77c-6257be58.mov"
converted_video_path = "/kaggle/working/test_video_480p.mp4"
os.system(f'ffmpeg -i "{input_video_path}" -vf scale=640:480
"{converted_video_path}"')
input_video_path = converted_video_path
```

```
def detect_edges(image):
    if image is None or image.size == 0:
        print("Warning: Empty image received for edge detection")
        return np.zeros((480, 640), dtype=np.uint8)
```

```
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    blurred = cv2.GaussianBlur(gray, (5, 5), 0)
    edges = cv2.Canny(blurred, 50, 150)
    return edges
```

```
def process_frame(frame):
    if frame is None or frame.size == 0:
        print("Warning: Empty frame detected, skipping...")
        return None
```

```
    edges = detect_edges(frame)
    results = model(frame)
    obstacle_detected = False
    for r in results:
        for box in r.bboxes:
            x1, y1, x2, y2 = box.xyxy[0]
            conf = box.conf[0]
            cv2.rectangle(frame, (int(x1), int(y1)), (int(x2),
int(y2)), (0, 255, 0), 2)
            cv2.putText(frame, f"Obstacle {conf:.2f}", (int(x1),
int(y1 - 5)), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0, 255, 0), 2)
            obstacle_detected = True
```

```
    combined = cv2.addWeighted(frame, 0.8, cv2.cvtColor(edges,
```

```

cv2.COLOR_GRAY2BGR), 0.5, 0)
lane_pixels = np.count_nonzero(edges)
if lane_pixels < 1000:
    navigation_command = "No Lane Detected - Stop!"
elif obstacle_detected:
    navigation_command = "Obstacle Ahead - Stop or Turn!"
elif lane_pixels > 4000:
    navigation_command = "Lane Curving - Adjust Steering"
else:
    navigation_command = "Move Forward"
cv2.putText(combined, navigation_command, (50, 50),
cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 0, 255), 2)
return combined

def process_video(input_video_path, output_video_path):
    cap = cv2.VideoCapture(input_video_path)
    fourcc = cv2.VideoWriter_fourcc(*"mp4v")
    fps = int(cap.get(cv2.CAP_PROP_FPS))
    frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
    frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
    out = cv2.VideoWriter(output_video_path, fourcc, fps,
(frame_width, frame_height))

    frame_limit = 300
    frame_count = 0
    while cap.isOpened() and frame_count < frame_limit:
        ret, frame = cap.read()
        if not ret or frame is None:
            print("Warning: Empty frame detected, skipping...")
            continue
        processed_frame = process_frame(frame)
        if processed_frame is None:
            print("Warning: Skipping frame due to processing error")
            continue
        out.write(processed_frame)
        frame_count += 1
    cap.release()
    out.release()

output_video_path = "/kaggle/working/output_video.mp4"
process_video(input_video_path, output_video_path)

```

0: 384x512 (no detections), 34.8ms
Speed: 3.4ms preprocess, 34.8ms inference, 28.9ms postprocess per
image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.6ms
Speed: 2.2ms preprocess, 6.6ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.9ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 2.0ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.7ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.5ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.4ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.4ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 2.0ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.5ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.5ms preprocess, 6.4ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.7ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.5ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms

Speed: 1.5ms preprocess, 6.5ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.3ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.5ms preprocess, 6.4ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.6ms

Speed: 1.6ms preprocess, 6.6ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.4ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.0ms
Speed: 1.5ms preprocess, 6.0ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.6ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.4ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.0ms
Speed: 1.3ms preprocess, 6.0ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.5ms preprocess, 6.2ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.7ms
Speed: 1.8ms preprocess, 6.7ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.5ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.3ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.3ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.7ms preprocess, 6.1ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.3ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.3ms preprocess, 6.2ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 7.6ms

Speed: 1.4ms preprocess, 7.6ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.8ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.3ms preprocess, 6.3ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.6ms
Speed: 1.5ms preprocess, 6.6ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.3ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.4ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.4ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.6ms

Speed: 1.6ms preprocess, 6.6ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.3ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.3ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.7ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.3ms preprocess, 6.3ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.5ms preprocess, 6.4ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.7ms preprocess, 6.3ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.4ms preprocess, 6.1ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms

Speed: 1.5ms preprocess, 6.5ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.7ms preprocess, 6.1ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.3ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.4ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.4ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.7ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.3ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.6ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.6ms preprocess, 6.2ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.6ms preprocess, 6.2ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.6ms preprocess, 6.4ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.5ms preprocess, 6.3ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.3ms preprocess, 6.4ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.4ms preprocess, 6.1ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.3ms preprocess, 6.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.3ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.6ms preprocess, 6.1ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.4ms preprocess, 6.4ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.4ms preprocess, 6.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.8ms

Speed: 1.7ms preprocess, 6.8ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.5ms preprocess, 6.4ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.7ms

Speed: 1.5ms preprocess, 6.7ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.3ms preprocess, 6.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.5ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.3ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.7ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.3ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.6ms preprocess, 6.1ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.4ms preprocess, 6.4ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image

at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.7ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.5ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.3ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.7ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.8ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.6ms preprocess, 6.2ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.4ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.7ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 7.3ms
Speed: 1.7ms preprocess, 7.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.5ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.4ms preprocess, 6.5ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.3ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 141.4ms postprocess per
image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.6ms
Speed: 2.0ms preprocess, 6.6ms inference, 0.6ms postprocess per image

at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.4ms

Speed: 1.8ms preprocess, 6.4ms inference, 1.2ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.9ms

Speed: 1.5ms preprocess, 6.9ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.6ms

Speed: 1.6ms preprocess, 6.6ms inference, 0.6ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.6ms

Speed: 1.6ms preprocess, 6.6ms inference, 1.2ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.4ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.3ms

Speed: 1.5ms preprocess, 6.3ms inference, 1.3ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.7ms preprocess, 6.2ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.4ms

Speed: 1.6ms preprocess, 6.4ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.4ms

Speed: 1.4ms preprocess, 6.4ms inference, 1.1ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.4ms

Speed: 1.5ms preprocess, 6.4ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.3ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.8ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.6ms
Speed: 1.7ms preprocess, 6.6ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.3ms
Speed: 1.7ms preprocess, 6.3ms inference, 1.3ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.4ms
Speed: 1.5ms preprocess, 6.4ms inference, 1.1ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.5ms
Speed: 1.7ms preprocess, 6.5ms inference, 1.2ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.3ms
Speed: 1.4ms preprocess, 6.3ms inference, 1.1ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.3ms
Speed: 1.3ms preprocess, 6.3ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.3ms
Speed: 1.5ms preprocess, 6.3ms inference, 1.1ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.8ms
Speed: 1.6ms preprocess, 6.8ms inference, 1.0ms postprocess per image

at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.2ms

Speed: 1.4ms preprocess, 6.2ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.3ms

Speed: 1.6ms preprocess, 6.3ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.3ms

Speed: 1.5ms preprocess, 6.3ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.3ms

Speed: 1.6ms preprocess, 6.3ms inference, 1.2ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.2ms

Speed: 1.6ms preprocess, 6.2ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.5ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.7ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.7ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.7ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.6ms

Speed: 1.6ms preprocess, 6.6ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.6ms

Speed: 1.7ms preprocess, 6.6ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.4ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.4ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 7.1ms
Speed: 1.6ms preprocess, 7.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.8ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.7ms
Speed: 1.6ms preprocess, 6.7ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.5ms preprocess, 6.4ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.4ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.5ms
Speed: 1.5ms preprocess, 6.5ms inference, 1.1ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.5ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.7ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image

at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.3ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.4ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.3ms preprocess, 6.4ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.7ms preprocess, 6.1ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.4ms preprocess, 6.4ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms

Speed: 1.7ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.8ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.4ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.6ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.6ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.6ms
Speed: 1.5ms preprocess, 6.6ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.6ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.4ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.7ms
Speed: 1.5ms preprocess, 6.7ms inference, 0.4ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.6ms preprocess, 6.4ms inference, 0.4ms postprocess per image

at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.7ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms

Speed: 1.7ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.6ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.3ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.3ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 7.0ms

Speed: 2.1ms preprocess, 7.0ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.6ms

Speed: 1.8ms preprocess, 6.6ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.6ms

Speed: 1.6ms preprocess, 6.6ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms

Speed: 1.5ms preprocess, 6.5ms inference, 0.6ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.5ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.5ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 1.2ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.1ms
Speed: 1.5ms preprocess, 6.1ms inference, 1.1ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.3ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.5ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.5ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.6ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.5ms
Speed: 1.6ms preprocess, 6.5ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.3ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.5ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.3ms preprocess, 6.2ms inference, 0.6ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.3ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.5ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.5ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms
Speed: 1.4ms preprocess, 6.1ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.7ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.7ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms
Speed: 1.7ms preprocess, 6.2ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.2ms
Speed: 1.4ms preprocess, 6.2ms inference, 1.2ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.3ms
Speed: 1.6ms preprocess, 6.3ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.3ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.6ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms

Speed: 1.5ms preprocess, 6.4ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.4ms preprocess, 6.2ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.4ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.2ms

Speed: 1.6ms preprocess, 6.2ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.3ms preprocess, 6.1ms inference, 0.4ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.1ms

Speed: 1.5ms preprocess, 6.1ms inference, 1.1ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.2ms

Speed: 1.5ms preprocess, 6.2ms inference, 1.0ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.4ms

Speed: 1.5ms preprocess, 6.4ms inference, 1.0ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.6ms

Speed: 1.5ms preprocess, 6.6ms inference, 1.0ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.1ms

Speed: 1.6ms preprocess, 6.1ms inference, 0.5ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.4ms

Speed: 1.5ms preprocess, 6.4ms inference, 1.0ms postprocess per image at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.4ms
Speed: 1.6ms preprocess, 6.4ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 (no detections), 6.4ms
Speed: 1.5ms preprocess, 6.4ms inference, 0.5ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.1ms
Speed: 1.6ms preprocess, 6.1ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 1 obstacle, 6.1ms
Speed: 1.4ms preprocess, 6.1ms inference, 1.1ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 2 obstacles, 6.3ms
Speed: 1.6ms preprocess, 6.3ms inference, 1.2ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 2 obstacles, 6.1ms
Speed: 1.4ms preprocess, 6.1ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 2 obstacles, 6.3ms
Speed: 1.6ms preprocess, 6.3ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 2 obstacles, 6.3ms
Speed: 1.5ms preprocess, 6.3ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 2 obstacles, 6.3ms
Speed: 1.5ms preprocess, 6.3ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

0: 384x512 2 obstacles, 6.1ms
Speed: 1.7ms preprocess, 6.1ms inference, 1.0ms postprocess per image
at shape (1, 3, 384, 512)

Video(output_video_path, embed=True)

<IPython.core.display.Video object>

```
output_video_path = "/kaggle/working/output_video.mp4"
zip_path = "/kaggle/working/output_video.zip"
shutil.make_archive(zip_path.replace(".zip", ""), 'zip',
"/kaggle/working", "output_video.mp4")
zip_path
```

```
'/kaggle/working/output_video.zip'
```